Sheltered Bidding: The Within-Auction Cost of SME Set-Asides*

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Abstract

Why do SME set-asides raise procurement prices? Two non-exclusive mechanisms operate: the bidder pool shrinks, and the auction clears at a different point in the surviving cost distribution. This paper estimates the total impact and decomposes it into these two channels. A March 2018 reversal extended an SME-only rule to medical supplies on São Paulo's centralized procurement platform; a difference-in-differences against 76 never-treated product groups recovers *a roughly 10 percent rise in winning prices* and a doubling of SME participation. An asymmetric IPV model identified from Pregão drop-out bids then separates the channels under observed equilibrium entry—a policy-relevant accounting decomposition identified by the auction format plus observed entry, not a full structural entry equilibrium. The within-auction component—*sheltered bidding*—accounts for *two-thirds to three-quarters of the simulated effect* across classes. The implied welfare cost reaches *R$ 55 million per year* on a single product group of São Paulo's R$ 13 billion platform. A 10% price preference welfare-dominates the set-aside in thick standardized markets at near-zero fiscal cost; the ranking turns conditional in thin pharmaceutical markets, where the equilibrium-selection treatment of the protected pool becomes first-order. The conditional ranking is the policy-design statement: bidder

* *This version: April 2026. I thank Paulo Furquim de Azevedo for early discussions, Sergio Firpo, Gustavo Torres, Gabriel Madeira, and Sandro Cabral for valuable comments, and participants at the Insper brown bag for helpful feedback. I also thank the BEC/SEFAZ-SP team for data access. All remaining errors are my own.*

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exclusion is hardest to defend where the protected pool is thick and the good standardized, and most defensible—if at all—where it is not.

Keywords: public procurement, SME set-asides, asymmetric auctions, structural estimation, welfare, marginal cost of public funds

JEL MSC: H32, H57, L26, L53, D44

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1. Introduction

Procurement set-asides for small firms are among the most widely deployed government-procurement instruments worldwide (Bosio et al., 2022). They were designed on a textbook intuition: bringing more bidders to a protected segment should reduce procurement prices. The intuition has not survived contact with the data. A decade of empirical evidence places the price cost of bidder-exclusion rules in the ten-to-fifteen percent range (Marion, 2007; Krasnokutskaya, 2011; Athey et al., 2013; Nakabayashi, 2013; Hyytinen et al., 2018). Two non-exclusive mechanisms operate behind that markup: the bidder pool shrinks, and the auction clears at a different point in the surviving cost distribution—an order-statistic shift the structural framework identifies separately from any active behavioral adjustment by the surviving bidders. Both matter for welfare and for the policy fix; the prescription depends on the relative weight of each channel, not on which alone wins. Price regressions identify the total but cannot disentangle the two, and the welfare ranking of bidder exclusion against alternative preference instruments turns on the decomposition—a ranking that, as this paper shows, is conditional on the thickness of the protected pool and the heterogeneity of the good rather than universal. On São Paulo's medical-supplies procurement alone, the structural exercise of this paper prices the rule at R$ 55–128 million (US$ 16–37 million) per year of welfare loss and decomposes that loss into the two channels.

A March 2018 administrative reversal in São Paulo state, Brazil, supplies the natural experiment. The state extended an existing SME-only procurement rule to medical and hospital supplies—the last major exception—on the country's largest centralized state-government procurement platform, the Bolsa Eletrônica de Compras (BEC). The reversal was a legal opinion internal to the state prosecutor's office, defended on isonomy grounds and uncorrelated with prices, competition, or supplier complaints in the affected product group; the timing was bureaucratic. A difference-in-differences against 76 never-treated product groups, reported as design validation in Appendix I, estimates that absolute winning prices rose by nearly 10 percent on the same window used by the structural exercise—squarely at the lower end of the Marion (2007)–Hyytinen et al. (2018) literature range—with SME participation roughly doubling. The DiD coefficient fixes the average effect on observed log prices; it does not say which mechanism produced the markup or what the rule costs in welfare terms. The structural exercise of this paper recovers both, anchored to the same magnitude.

This paper provides that structure. The object is not a full structural entry equilibrium; it is the policy-relevant decomposition identified by the auction format plus observed equilibrium entry, in the sense made precise by Sections 4–5.[1] I specify an asymmetric independent-private-values reading of the platform's Pregão English-reverse auction format. Bidders draw costs from type-specific distributions F_c^{SME} and $F_c^{\neg}$ and compete under a descending clock; losers exit at cost and the last firm supplies the contract. Drop-out prices of losers therefore point-identify F_c^k under Haile and Tamer (2003), without inverting a bid function or imposing parametric structure on bidding. An auction-level common scale is shrunk out following Krasnokutskaya (2011); winners are treated as

[1]Three magnitudes appear in this paper and should not be conflated. The *DiD coefficient* (10–11 percent) is the average effect on observed log winning prices net of item and month fixed effects. The *structural counterfactual* (+0.231 in non-pharma, +0.338 in pharma) is the simulated mean of the second-order statistic under each counterfactual entry profile, scaled by the reference price; in p_{S_1} units it is three to four times the DiD coefficient because the two objects average over different primitives. The *welfare cost* (28.9 percent of p_{S_1} in non-pharma, 44.8 percent in pharma at $\lambda = 0.30$) adds the MCPF distortion on top of the allocative wedge.

left-censored at the second-lowest exit and recovered by a Turnbull NPMLE robustness. Entry is handled in the Athey et al. (2011) spirit: pre- and post-period bidder counts are taken as equilibrium arrival rates, and the main specification allows the post-policy SME cost distribution to differ from its pre-policy counterpart as part of the equilibrium-selection object. The exercise is conducted under observed equilibrium entry; it is not a fully solved nested entry-bidding equilibrium, and the decomposition that follows is a counterfactual-simulation accounting under the modeling assumptions stated.

Three findings organize the paper.

(i) The price effect operates predominantly within the auction—*sheltered bidding*. Let p_{S_1} be the simulated winning price under the pre-policy open regime, p_{S_2} the SME-only counterfactual at the pre-policy SME pool, and p_{S_3} the SME-only counterfactual at the observed post-policy SME pool. In the main specification, $p_{S_3} - p_{S_1} = +0.231$ of the reference price in non-pharma and +0.338 in pharma (Table D.21). The within-auction piece—which I label *sheltered bidding*—is 74.0 percent of that effect in non-pharma and 66.1 percent in pharma. The share stays in the 73–85 percent range across cost-distribution estimators (Turnbull NPMLE: 74.0, 82.0) and across the two specifications of the post-policy SME pool (strict invariance: 85, 79). The set-aside's price cost is mostly a property of which order statistic the restricted pool generates, not of how many firms it contains.

(ii) Endogenous SME entry attenuates a 59 percent larger latent shock in non-pharmaceuticals and an 81 percent larger one in pharmaceuticals. Hold the SME pool at its pre-policy size while imposing the SME-only rule and the simulated price effect rises to 159 percent of the realized one in non-pharma and 181 percent in pharma. Without endogenous entry the set-aside would cost 59 percent more in price and welfare in non-pharmaceuticals and 81 percent more in pharmaceuticals. Participation is the government's partial offset, not the source of the markup: the SMEs the protected segment induces to enter absorb part of the within-auction shock, though not all of it.

(iii) A 10% price preference welfare-dominates the set-aside at near-zero fiscal cost in thick standardized markets; pharma bifurcates. In thick standardized markets the simulated price shifts by -0.004 of the reference price under the preference, against $+0.231$ under the set-aside. The dominance holds under both the main equilibrium-selection and the strict-primitive-invariance treatment of the post-policy SME pool. Scaled to the post-policy run-rate of Group-65 procurement on São Paulo's R$ 13 billion centralized platform, the implied annual welfare loss of the set-aside is on the order of R$ 55–128 million (US$ 16–37 million). In thin heterogeneous pharmaceutical markets the comparison bifurcates: the preference dominates the set-aside under the main equilibrium-selection reading, but under strict primitive invariance the ranking flips, with the implicit welfare weight required to prefer the set-aside falling to 0.7. The bifurcation is the diagnostic: it pins down where the modeling treatment of the protected pool is first-order, and it is exactly where the policy ranking demands caution.

Three contributions follow. *First*, a within-auction structural decomposition of an SME procurement preference identified from a single legal trigger, on a combination of conditions rare in the literature: a total-exclusion legal trigger (clean regime separation), a Pregão drop-out format (point-identification of cost distributions), an auction-level UH correction (Krasnokutskaya, 2011), observed-equilibrium-entry treatment à la Athey et al. (2011), and a Saez and Stantcheva (2016) welfare-weight identity. Marion (2007) estimates the price effect of federal SME set-asides via reduced-form variation; Krasnokutskaya (2011) decomposes California highway bid preferences via cross-equation constraints, and Athey et al. (2013) prices the efficiency-vs-redistribution trade-off between set-asides and subsidies in U.S. timber. This paper extends the second approach to a total-exclusion rule on a single clean legal trigger, computing the welfare ranking against alternative instruments within the same framework and identifying the market-structure conditions under which the standard preference-over-exclusion ranking does not transport. *Second*, endogenous SME entry is part of the welfare object, not a side issue: a fixed-pool counterfactual misstates the realized policy cost by roughly half. *Third*, the welfare comparison is a market-design statement, not a universal ranking—the 10% preference welfare-dominates in thick standardized markets across both specifications, while pharma's bifurcation identifies precisely the market conditions under which the modeling treatment of the protected pool is first-order.

The paper sits in dialogue with three lines of empirical work: recent procurement-design papers on emerging-market public contracting (Decarolis et al., 2025; Coviello et al., 2026; Ferraz et al., 2016; Szerman, 2023), the political economy of public contracting (Colonnelli and Prem, 2022), and the public-finance evaluation of government programs in developing economies (Best et al., 2023; Bandiera et al., 2021; Olken, 2007).

A pair of comparable furosemide 40 mg purchases by the same São Paulo public buyer eight months apart and bracketing the cutoff fixes the magnitudes. In February 2018 the unit cleared roughly 12 percent below the buyer's reference price; in October 2018 it cleared just above. The difference is one realization of the average direction the design estimates across the 297,967 firm-auction observations and 97,993 distinct auctions of the 18-month structural sample on each side of the March 2018 cutoff—10–11 percent on the absolute final price in the difference-in-differences benchmark of Appendix I, the magnitude the structural exercise of this paper interprets and prices. Table 1 summarizes the headline numbers across the four reporting layers used throughout the paper. *The rows are not alternative estimates of the same object*: the DiD coefficient, the structural counterfactual, the within-auction share, and the welfare cost are distinct objects identified at distinct levels of the framework, related by the mechanical bridge developed in Section 6.4 and the welfare-weight identity of Section 9.

Table 1: Headline magnitudes.

	Non-pharma	Pharma
DiD on $\log p^{\text{final}}$ (absolute, App. Appendix I)	+0.10 to +0.11	
DiD on $p^{\text{final}}/p^{\text{ref}}$ (App. Appendix I)	+0.06	
Simulated $p_{S_3} - p_{S_1}$, units of p^{ref}	+0.231	+0.338
Within-auction share, main specification	74.0%	66.1%
Within-auction share, Turnbull NPMLE	74.0%	82.0%
Within-auction share, strict primitive invariance	85.0%	79.0%
Welfare cost (% of p_{S_1}, $\lambda = 0.30$)	28.9%	44.8%
$w_\star^{\text{SME}}$ to prefer set-aside, main spec.	2.42	2.61
$w_\star^{\text{SME}}$, strict invariance	>1	0.7

DiD magnitudes from the difference-in-differences benchmark in Appendix I (against 76 never-treated product groups, matched window of 18 months on each side of the March 2018 cutoff). Structural magnitudes from the asymmetric IPV counterfactual under observed equilibrium entry; sources by row: Sec. 6 (decomposition), Sec. 8 (Turnbull NPMLE; strict invariance), Sec. 7 (welfare cost), Sec. 9 (welfare weights). The pharmaceutical welfare-dominance ranking against a 10% price preference is robust under the main specification but reverses under strict primitive invariance ($w_\star^{\text{SME}} = 0.7 < 1$); the non-pharmaceutical ranking is preserved under both specifications.

2. Institutional Background

The identifying variation in this paper is institutional rather than economic: a generic legal reinterpretation, transversal to product categories, mechanically unlocked Group 65 from a procurement default that had bound it for reasons unrelated to medical-supplies market conditions. Three features of the reform support that reading. The trigger is a legal opinion at the São Paulo state Attorney General's office (PGE-SP), not a Group-65 policy initiative. The mechanism is the interaction of the opinion's general re-reading of the SME-only threshold with the typical contract structure of Group-65 notices. The audit court (TCE-SP) ratified the reinterpretation five months after the empirical cutoff, confirming the operational sequence rather than initiating it. Table 2 reconstructs the chronology; the rest of this section develops the three features in turn.

Brazil's federal SME statute (Complementary Law 123/2006, amended for procurement by LC 147/2014 and for revenue brackets by LC 155/2016) defines micro and small enterprises by annual gross revenue ceilings of R\$360,000 (US\$103,000) and R\$4,800,000

(US$1.37 million), respectively, and mandates exclusive SME tenders for items valued at R$80,000 or less. LC 147/2014 converted the SME-only rule from optional ("may") to mandatory ("must"), with departures requiring a costly justification to the audit courts; state-level adherence to SME-only tendering for eligible items rose from roughly 13% to nearly 70% in the years that followed.[2]

The state of São Paulo has operated the Bolsa Eletrônica de Compras (BEC) since September 2000 (Decree 45.085/2000), and the Pregão electronic reverse-auction format that anchors this paper's analytical sample was regulated within BEC by Decree 49.722/2005. BEC centralizes Pregão for standardized-goods procurement by state bureaus and affiliated units; the structural sample covers 4.8 million item-level transactions across all product groups over the 36-month estimation window, of which Group 65 accounts for roughly 27 percent. Its catalog is organized in three tiers—group, class, and item—so that "medical, dental, and hospital equipment and supplies" form Group 65, within which class 6531 identifies medications subject to price ceilings set by the Câmara de Regulação do Mercado de Medicamentos (CMED).[3]

Group 65 was a singular exception to the 2014 rule, but for a transversal legal reason rather than a sector-specific one. Until late 2017, the prevailing interpretation of art. 48, I of LC 123/2006 was that the R$80,000 threshold for SME-only tenders applied to the *total estimated value of the procurement notice*, not to each individual item or lot—a reading sustained by TCE-SP in eTC-5509.989.15 (23 September 2015, by tie-breaking

[2]Adherence figures and Group-65 sub-rates are author tallies from the BEC microdata extract used in the structural sample (Section 3; replication numbers in Table B.8), restricted to items satisfying the federal R$80,000 threshold. The R$1.37M EPP ceiling raised by LC 155/2016 has no direct bite on the structural sample, where SME status is identified from the BEC *enquadramento* flag and the Receita Federal *porte* register at the firm-auction level; the bracket change is reported here for institutional completeness.

[3]Brazilian institutional terms used throughout the paper (Pregão, Convite, BEC, CADMAT, CMED, PGE-SP, TCE-SP, Sefaz-SP) are defined in Appendix A. Platform institutional description at `www.bec.sp.gov.br/BECSP/Quem_Somos/Quem_Somos_BEC_novo.aspx` (retrieved April 2026); the 4.8-million item-level count and the 860,000 purchase-offer count are author tallies from the same BEC extract used in the structural analysis.

vote). Group-65 procurement notices systematically combine small individual items (medications, supplies, dressings) with large notice totals reflecting bulk public-sector purchasing, so the totalizing reading mechanically barred Group 65 from the SME-only default—not because medical supplies were classified as "strategic," but because their typical procurement structure was exactly what the threshold-on-total rule excluded.

On 11 December 2017, PGE-SP issued **Parecer Sub-G Cons. nº 151/2017**, holding that the R\$80,000 threshold applies item-by-item—allowing mixed-exclusivity procurement notices in which items below R\$80,000 are SME-only and items above remain open—and that the more generous federal regime introduced by LC 147/2014 supersedes the previously applicable São Paulo state acts (Lei Estadual 13.122/2008 and Decree 54.229/2009) that had embedded the restrictive interpretation in state-level practice.[4][5]

Operationally, BEC moved before PGE consolidated the legal interpretation. COMUNICADO BEC nº 02/2017 (18 July 2017) introduced the SME-only OC functionality, and COMUNICADO BEC nº 03/2017 (20 November 2017) expanded it to mixed-exclusivity Pregão notices combining ample-participation, 25%-quota, and SME-only items in the same notice—the precise operational mechanism that Parecer 151/2017 ratified three weeks later.[6] The empirical cutoff of March 2018 therefore captures the moment of mass take-up of the new functionality by Group-65 PBUs rather than formal regulatory enable-

[4]Parecer Sub-G Cons. nº 151/2017, Subprocuradoria-Geral de Consultoria da PGE-SP. Headnote (author's translation from Portuguese): *"Tenders exclusive to micro and small enterprises (art. 48, I, LC nº 123/2006) must be conducted for items or lots whose individually estimated value does not exceed the legal ceiling of R\$80,000.00,* ***regardless of the total value of the Purchase Offer****. . . . Full applicability of the federal regulation."*

[5]Disclosed at the BEC/SP legislation portal (https://www.bec.sp.gov.br/becsp/Legislacao/UI_Selecao.aspx, *Pareceres* section, retrieved April 2026); the Boletim CEPGE indexes only Pareceres da Procuradoria Administrativa, so the BEC portal is the canonical channel for Sub-G Cons. pareceres.

[6]Both Comunicados BEC are disclosed at the BEC/SP legislation portal cited above (*Comunicados* section). TCE-SP itself reversed its restrictive reading on 16 May 2018 in eTC-9589.989.18 (Tribunal Pleno), aligning with PGE-SP's expansive interpretation two and a half months after the empirical cutoff of this paper. Both TCE-SP rulings are accessible at the court's jurisprudence portal (https://www.tce.sp.gov.br/jurisprudencia).

ment (which was incremental between July and November 2017); SME-only adherence within Group 65 jumped to 43% on average in the months following the cutoff, below the rate for other groups largely because class-6531 pharmaceuticals are oligopolistic and often fail the three-eligible-SMEs requirement.

What matters for identification is that the trigger was a generic statutory reinterpretation, and the unlocking of Group-65 set-asides was a mechanical consequence of how medical-supplies notices are structured. PGE-SP's parecer made no substantive claim about medical supplies or any other product group; it pronounced on a general legal question, with timing set by the bureaucratic process inside PGE-SP and by when BEC operationalized the new interpretation. The TCE-SP reversal in May 2018 lies after the cutoff and confirms that the trigger was a top-down legal interpretation: when the audit court itself flipped, it ratified what PGE-SP and BEC had already established for state-level procurement rather than initiating an independent change in the Group-65 environment. The change is therefore exogenous to Group-65 market conditions on the dimensions the design tests. This is the institutional basis for Assumption 3: the cutoff shifts admissibility and equilibrium entry selection, not the underlying production technology of the goods themselves. Three empirical checks corroborate the institutional reading: the reduced-form DiD against the 76 never-treated product groups (Appendix I); the primitive-invariance KS test on non-SME F_c (Table C.12); and the cross-modality consistency between Convite GPV inversion and Pregão drop-out recovery in the unaffected pharma non-SME Pre stratum (Figure 1).

3. Data and Sample Construction

The structural exercise rests on a universe-coverage administrative panel: every bid submitted on every BEC procurement auction over a four-year window, each with the firm's identifier and its registry-validated SME status. The raw extract carries 3.7 million

Table 2: Institutional timeline—federal and São Paulo state acts relevant to the structural sample.

Date	Act	Content
Sep 2000	Decree 45.085/2000 (SP)	BEC platform created; renamed Decree 45.695/2001.
Jun 2005	Decree 49.722/2005 (SP)	Pregão electronic reverse-auction regulated within BEC.
Dec 2006	LC 123/2006 (federal)	SME statute (*Estatuto Nacional*).
2008	Lei Estadual 13.122/2008 (SP)	SP state-level SME procurement regulation (less favorable than federal post-2014).
2009	Decree 54.229/2009 (SP)	Regulates Lei Estadual 13.122/2008.
Aug 2014	LC 147/2014 (federal)	SME-only default (“may”→“must”); R$80,000 threshold.
Sep 2015	TCE-SP eTC-5509.989.15	Restrictive reading of art. 48, I (R$80,000 on notice total) sustained by tie-breaking vote.
Oct 2016	LC 155/2016 (federal)	EPP revenue ceiling R$3.6M→R$1.37M (effective Jan 2018).
18 Jul 2017	COMUNICADO BEC nº 02/2017	SME-only OC functionality enabled in BEC.
20 Nov 2017	COMUNICADO BEC nº 03/2017	Mixed-exclusivity (Ample / 25% quota / SME-only) item-by-item enabled.
11 Dec 2017	PGE-SP Parecer Sub-G Cons. 151/2017	Item-by-item R$80,000 reading; federal regime supersedes Lei 13.122/2008.
March 2018	Mass adoption (empirical cutoff)[a]	Group-65 PBU take-up of the new functionality (first wave of adopting PBUs).
16 May 2018	TCE-SP eTC-9589.989.18	Tribunal Pleno reverses, aligns with PGE-SP expansive reading.
Apr 2021	14.133/2021 (federal)	New procurement statute; SME defaults preserved.

[a] BEC formally enabled the new functionality on 18 July 2017 (Comunicado 02/2017) and expanded it to item-by-item mixed exclusivity on 20 November 2017 (Comunicado 03/2017). The empirical cutoff in March 2018 (BEC purchase-order sequence index data_oc_numb = 698) corresponds to the moment of mass take-up by Group-65 PBUs, not formal enablement; the gap reflects the diffusion lag between operational availability and PBU adoption. The TCE-SP reversal in May 2018 (eTC-9589.989.18) post-dates the cutoff, confirming that the trigger of the design is the PGE-SP/BEC operationalization sequence, not the audit-court realignment.

bid-level observations across 82,569 items, 832,984 purchase orders, and 1,344 public buyer units—a 36-month subset of the 4.8 million item-level transactions BEC intermediates over the platform throughput cited in Section 2. Group 65 (medical, dental, and hospital supplies) accounts for roughly 27 percent of transactions over the window and is the treated group; the remaining 76 product groups form the never-treated control set in the difference-in-differences benchmark of Appendix I.

The structural analysis narrows to a sharper sample. Identification uses a symmetric window of 18 months on each side of the March 2018 cutoff (36 months in total, the "18m" label of Table E.30); in Stata monthly-date units the window is $[680, 715]$, or September 2016 through August 2019 inclusive. Within that window, I restrict to Pregão auctions in Group 65. The restriction does two things. Pregão is the descending-clock format that admits point identification of cost distributions from drop-out bids, so the structural estimation requires it; Group 65, in turn, is where the policy bound—the only product group to switch regime during the sample.

Bids are organized at the firm-auction level—one observation per (firm, item-auction) pair with a winner/loser flag. Three filters discipline the structural sample without re-shaping it. The bid filter $c_\epsilon = b/p^{\text{ref}} \in (0, 3]$ removes implausible outliers (under 0.4 percent of bids, almost always data-entry errors at the hundred-times-reference level). The participation filter $n \geq 2$ keeps only auctions where point identification of F_c and the BNE counterfactual are both well-defined. The classification step attaches a type flag (SME or non-SME, from the BEC administrative registry validated against historical bidding to absorb transient status changes) and a pharma flag (CADMAT classes 6531, 6532, 6536, 6581—CMED-regulated pharmaceuticals and biologicals—are pharma-narrow; the remainder of Group 65 is non-pharma). The resulting structural sample carries 297,967 firm-auction observations across 97,993 distinct auctions, balanced symmetrically across Pre (September 2016–February 2018, n = 48,740 auctions) and

Post (March 2018–August 2019, $n = 49{,}253$). Stratum-level descriptive counts appear in Appendix Tables B.9 and B.10.

Auction-level unobserved heterogeneity is material and not fully absorbed by the reference-price normalization. Two items inside the same CADMAT class can share a scale component that moves every bid in the auction together—volume differences, ANVISA registration burden, contract-specific complexity. I handle this with the Krasnokutskaya (2011) method-of-moments variance decomposition followed by best-linear-predictor shrinkage; details in Section 5, with the recovered intraclass correlations and the variance panel in Appendix Table C.14. The cleaned residuals $c_{\epsilon}^{\text{clean}} = \exp(\hat{e}_{it})$ enter all downstream structural estimation.

Entry counts enter separately. For each auction I count the distinct firms of each type that submitted at least one bid, then average within (period $\times$ pharma) cells. Entry rates enter the BNE as observed equilibrium arrivals (Section 5); the panel is Appendix Table D.19, with magnitudes interpreted in Section 6.

The analysis relies on four ancillary inputs. Reference prices come from the BEC catalog (`preco_ref`) and are used both for normalization and for pricing out the fiscal cost of the rule in reais. The reference price for each item is set ex ante by the buyer from the BEC catalog and does not move with the regime change: item-level fixed effects in the DiD specifications and the structural normalization absorb cross-sectional variation in p^{ref} across items, and the BEC catalog architecture sets p^{ref} at the item level for the entire window. The variable that changes around the cutoff is regime admissibility, not item-level catalog pricing. CADMAT class codes come from the BEC product hierarchy. The historical SME flag is reconstructed from firm-level bidding histories over the full 2014–2019 period to protect against transient status changes. The pharma flag follows the CADMAT class list above.

The pair of furosemide 40 mg purchases mentioned in the introduction (item 110639, a

CMED-regulated generic antihypertensive) sits in the institutional sample. February 2018: 10 firms bid, clearing price 12 percent below reference. October 2018: 3 SMEs bid, clearing price just above reference. Welfare arithmetic on this pair appears in Section 7; the structural averages are over 59,997 Pregão pharmaceutical auctions in the 18-month sample (Table B.9).

4. Model

A stylized model captures what the set-aside does and, as important, what it does not do. N potential bidders compete for a single indivisible contract. Each bidder i has type $k_i \in \{\text{SME}, \neg\}$ and draws a private cost of supply c_i from a type-specific distribution F_c^k, with F_c^{SME} and $F_c^{\neg}$ differing in ways the data are allowed to determine.

The auction format is the identifying payload. BEC implements procurement through a Pregão—a descending-clock electronic English-reverse auction, distinct both from first-price sealed-bid designs (Convite, in the smaller-value Brazilian segment) and from the Dutch descending-price format more commonly studied in the auction literature. The clock opens at the buyer's reference price p^{ref}, firms post downward-revisable offers until only one firm remains, and that firm supplies the contract at the clearing price. What makes this format analytically convenient is a standard result of Haile and Tamer (2003): under independent private values, the weakly dominant strategy is to exit the auction when the running clock price reaches one's own cost. Drop-out prices of losers therefore *point-identify* their costs, without requiring the researcher to invert a bid function or assume a parametric form for the bidding strategy. The winner, who never drops out before the clock stops, leaves only an upper bound $c_{\text{win}} \leq c_{(2)}$ (the cost of the second-to-last firm to exit)—and that upper bound is exactly what the Turnbull robustness check in Section 8 uses to deliver a non-parametric MLE of F_c^k. This point-identification under Pregão is the reason a structural analysis is available here where the cross-sectional first-price-sealed-bid settings of Marion (2007), Krasnokutskaya (2011), and Athey et al. (2011) have relied on cross-equation constraints to disentangle channels.

Why this framework. The choice of asymmetric IPV with the Krasnokutskaya UH correction and Athey–Seira reduced-form entry is dictated by the question and by what the BEC-Pregão data identify. Three alternatives merit explicit comment. *Common-value or affiliated-private-value formulations* (Milgrom and Weber, 1982; Hong and Shum, 2003) are most natural where bidders share an unknown component of the contract's value—e.g., construction contracts with incompletely observed specifications. The CADMAT-class restriction strips cross-item heterogeneity at the catalog level, and the cross-modality consistency check tolerates within-auction cost correlation up to $\rho_c \leq 0.3$ without rejecting the IPV restriction (Section 8). *Nested entry-bidding equilibria* à la Krasnokutskaya and Seim (2011) recover entry costs structurally from a free-entry indifference condition; the deliberately weaker reduced-form entry adopted here—Pre/Post bidder counts as observed equilibrium primitives in the Athey et al. (2011) spirit—keeps the welfare statement anchored to the legal trigger rather than to a participation-cost identity the trigger does not directly identify. *Open-format or FPSB equivalents* are bounded explicitly under Maskin and Riley (2000) in Section 8.4, with the V3-vs-V0 ranking preserved at the upper bound. The framework is therefore the lightest specification that sustains the within-auction-vs-entry decomposition while leaving alternative readings to the robustness battery.

The model relies on three substantive restrictions, stated formally below and defended in turn with the diagnostic evidence in Section 5.

Assumption 1 (Independent Private Values, asymmetric across types). *Costs c_i are drawn independently across bidders. Conditional on type k, $c_i \sim F_c^k$ with density f_c^k on support $[\underline{c}, \overline{c}] \subset \mathbb{R}_+$. The SME and non-SME distributions are not constrained to coincide.*

Assumption 1 permits but does not impose stochastic dominance between types; the empirical median-cost gap of 0.10–0.30 of the reference price across (pharma $\times$ type) cells (Section 5) supports asymmetry directly, and symmetric IPV would require this gap to vanish.

Assumption 2 (Multiplicative auction-level heterogeneity). *The normalized log-bid decomposes as $y_{it} = \log(b_{it}/p_t^{\text{ref}}) = a_t + e_{it}$, where a_t is an auction-level scale common to all bidders in auction t, and e_{it} is bidder-specific and independent across bidders within t. Both a_t and e_{it} are mean-zero with finite variances $\sigma_a^{2,k}$ and $\sigma_e^{2,k}$ that may vary by stratum.*

Assumption 2 is the Krasnokutskaya (2011) restriction separating a common scale from bidder-specific private values; multiplicativity is the natural form when item complexity or ANVISA burden scales costs uniformly across firms. The Turnbull NPMLE of Section 8 dispenses with Assumption 2 altogether and corroborates the baseline within 16 percent of the total effect.

Assumption 3 (Set-aside, with equilibrium SME selection). *Under the open regime both SMEs and non-SMEs enter with strictly positive probability. Under the SME-only regime, only SMEs are admissible. Within a short window around the cutoff, the underlying technology of supply is stable, but the equilibrium distribution of active SMEs under the restricted regime may differ from its pre-regime counterpart because the policy changes which SMEs find entry profitable. Write these distributions $F_c^{SME,\tau}$ for $\tau \in \{Pre, Post\}$.*

Proposition 1 (Identification of F_c^k under ascending-clock IPV). *Under Assumptions 1–3, drop-out exit prices in the Pregão English-reverse format point-identify the type-specific cost distribution F_c^k (Haile and Tamer, 2003), up to the auction-level scale a_t recovered by the variance decomposition in Assumption 2 (Krasnokutskaya, 2011). The post-policy SME distribution $F_c^{SME,Post}$ is identified as an equilibrium-selection object, with strict invariance ($F_c^{SME,Post} = F_c^{SME,Pre}$) recovered as a nested benchmark.*

Assumption 3 is the structural counterpart of the parallel-trends assumption in reduced-form DiD, with equilibrium selection made explicit à la Athey et al. (2011): the policy changes the admissible set and therefore the composition of SME bidders who enter, so $F_c^{\text{SME,Post}}$ is an equilibrium object rather than evidence of a primitive technological break. The primitive-invariance KS test on non-SME F_c (Table C.12) rejects strict invariance in pharma Pregão ($D = 0.0722$) and non-pharma Convite ($D = 0.0514$) at the operational threshold $D < 0.05$, and fails to reject in the other two strata. The cross-modality check (Table C.13) provides an additional consistency test.

Strict invariance as a limiting benchmark. A stronger reading would impose $F_c^{\mathrm{SME,Post}} = F_c^{\mathrm{SME,Pre}}$ and attribute the entire policy response to bidder counts alone. The KS tests in Tables C.12 and C.16 reject strict invariance of non-SME F_c in three of the four Pregão strata after UH cleaning (KS distances of 0.0613 non-pharma and 0.1407 pharma), so a structural exercise that imposed strict invariance on data that visibly reject it would be misspecified by construction. The main specification adopts equilibrium selection (Assumption 3) and reports strict invariance as a limiting bound in Section 8: the latter preserves the intensive-margin headline but changes the pharma welfare comparison because it shuts down equilibrium selection in who enters. The rejection is therefore evidence in favor of the modeling choice, not against it.

Equilibrium, entry, and counterfactuals. The auction format is interpreted as ascending-clock IPV: under Assumption 1, exit-at-cost is weakly dominant, the surviving bidder pays the second-lowest exit, and the expected transaction price equals $\mathbb{E}[c_{(2)}]$. Revenue equivalence (Vickrey, 1961; Maskin and Riley, 2000) extends the same object to FPSB; Krasnokutskaya (2011) provides the analog argument used here. The Bayes–Nash counterfactual reduces to Monte Carlo simulation over draws from the estimated F_c^k: stratum-specific bidder counts from the type-specific Poisson arrival process, costs from F_c^k, sorted to read off $c_{(1)}$ (production cost) and $c_{(2)}$ (expected transaction price). Equilibrium arrival rates enter as primitives, not derived from free entry: observed Pre counts $(N_{\mathrm{Pre}}^{\mathrm{SME}}, N_{\mathrm{Pre}}^{\neg})$ are status-quo arrivals; Post counts $(N_{\mathrm{Post}}^{\mathrm{SME}}, 0)$ are equilibrium arrival under the restricted regime, in the Athey et al. (2011) spirit. The implied zero-profit entry costs (Table D.20, R\$0.11–R\$2.46 per bidder) are per-auction calibration margins, not jointly identified fixed costs of the kind in Krasnokutskaya and Seim (2011); the decomposition that follows is counterfactual-simulation accounting under the maintained assumptions, and the welfare-cost expression of Section 7 prices the simulated price effect at the MCPF λ.[7]

The decomposition. Three counterfactual scenarios anchor the exercise:

- S_1: open regime at the pre-period pool. The baseline.
- S_2: SME-only at the pre-period SME pool. The non-SME type is removed; the SME pool is held at its observed pre-policy arrival rate. Within-auction component isolated.
- S_3: SME-only at the post-period SME pool. Uses the observed post-policy arrival rate and the post-policy SME cost distribution. Full policy effect with endogenous entry.

The simulated total effect on the second-order statistic decomposes arithmetically as

$$\underbrace{p_{S_3} - p_{S_1}}_{\text{simulated total effect}} = \underbrace{(p_{S_2} - p_{S_1})}_{\text{within-auction component (sheltered bidding)}} + \underbrace{(p_{S_3} - p_{S_2})}_{\text{participation-margin component}} . \quad (1)$$

Both components are properties of the simulation given F_c^k and the observed entry pattern; they sum to the simulated total effect by construction. The within-auction component can exceed the realized total effect when the participation-margin component partially offsets it, which is what the data deliver in non-pharmaceutical markets.

Sheltered bidding as a characterized property. The within-auction component can be characterized more generally within asymmetric-IPV auctions: any restriction

[7] Independence-across-auctions is assumed at the level of cost draws conditional on type and stratum. Within-auction correlation enters through the auction-level scale a_t (Assumption 2); the cluster bootstrap of Section 6 resamples at the auction level to preserve this correlation in the standard errors.

of admissibility to a single bidder type, holding the on-type pool fixed, generates an order-statistic shift with empirical content that survives outside the BEC-SP application.

Definition 1 (Sheltered bidding). *Consider an asymmetric IPV auction in which the buyer restricts admissibility to bidder type k. Let p_{open} denote the expected transaction price under the open regime with both types entering at observed equilibrium arrival rates, and let $p_{shelter}$ denote the expected transaction price under the type-k-only regime at the same on-type pool size as the open regime. The* sheltered-bidding component *of the policy effect is the equilibrium price shift*

$$\Delta_{shelter} \;\equiv\; p_{shelter} - p_{open},$$

identified separately from the participation-margin component $\Delta_{entry} \equiv p_{endo} - p_{shelter}$ *that captures the equilibrium response of type-k entry to the restriction. In the BEC-SP notation, $\Delta_{shelter} = p_{S_2} - p_{S_1}$ and $\Delta_{entry} = p_{S_3} - p_{S_2}$.*

Under Assumptions 1–3, Δ_{shelter} is point-identified from drop-out exits via Proposition 1; the expected second-order statistic shifts mechanically as a function of which order statistic is realized at each pool composition, without any active behavioral adjustment by surviving bidders (under ascending-clock IPV, exit-at-cost remains weakly dominant regardless of pool size). The component is structural in the sense that any policy intervention that removes a bidder type while holding the surviving pool fixed will produce a Δ_{shelter} predictable from F_c^k and $F_c^{\neg}$ alone—procurement set-asides, exclusionary clearance rules, supplier disqualification regimes, or any restriction that operates on the admissibility margin.

When sheltered bidding dominates. The decomposition predicts $|\Delta_{\text{shelter}}| > |\Delta_{\text{entry}}|$ when two conditions hold jointly: the off-type pool was thick enough in the open regime to materially shift the second-order statistic ($F_c^{\neg}$ contributes with non-negligible mass to $\mathbb{E}[c_{(2)}]$), and the type-k entry response to the restriction is bounded relative to the off-type removal effect. Both conditions are satisfied in the thick standardized non-pharmaceutical case of the BEC-SP data, where $\Delta_{\text{shelter}}/\Delta_{\text{total}} \approx 74.0$ percent. The pharmaceutical case sits closer to the boundary on both margins: a thinner off-type pool reduces $|\Delta_{\text{shelter}}|$, and the equilibrium-selection mechanism in $F_c^{\text{SME,Post}}$ introduces an additional source of variation in the entry margin that the static framework does not bound. The bifurcation between the model-stable non-pharmaceutical decomposition and the model-sensitive pharmaceutical one is therefore a prediction of the framework, not an artifact of the application, and the framework provides the diagnostic policy-makers need when choosing between bidder-exclusion and price-preference instruments across heterogeneous procurement categories.

5. Identification and Estimation

Three objects require estimation: the type-specific cost distributions F_c^k, the auction-level scale a_t, and the equilibrium entry counts. The third is directly observed. The first two require structure—F_c^k from the design of the Pregão auction (Haile and Tamer, 2003), a_t from the variance decomposition in Assumption 2 (Krasnokutskaya, 2011). Two diagnostic checks discipline the identification: a cross-modality consistency test that compares the Convite GPV recovery against the Pregão drop-out recovery in a stratum unaffected by the policy, and a Turnbull NPMLE that brackets the all-bidders estimator from below by removing the winner-censoring bias. Both are designed to fail visibly if the identifying assumptions are violated; both are passed.

5.1. Type-specific cost distributions

Under Assumption 1 and the English-reverse interpretation, the drop-out price of each loser is a point observation of that loser's cost. The main-text estimator is an *all-bidders* ECDF that adds the winner's final clock price to the stratum-specific ECDF; in a descending-clock auction this final price is approximately $c_{(2)}$, a tighter upper bound on the winner's cost than the zero imputed by losers-only omission. Two alternatives bracket all-bidders: losers-only overstates F_c^k in the left tail by omitting the lowest cost draw per auction; the Turnbull NPMLE (Section 8) treats the winner as left-censored at $c_{(2)}$ and delivers point identification, with the strict ordering $F_c^{\text{Turnbull}} \leq F_c^{\text{all-bidders}} \leq F_c^{\text{losers-only}}$ in the left tail. Across the eight (class $\times$ type $\times$ period) strata (Table E.24), the ordering holds at the median in all 8 strata and at the 75th percentile in 6 of 8 (the two pharmaceutical non-SME cells reverse by 0.01–0.02 of the reference price, within the bootstrap CI). The BNE decomposition computed on the Turnbull input differs from the all-bidders baseline by at most 16 percent of the total effect (Table E.25); losers-only is the input the DiD benchmark of Appendix I effectively absorbs through the item fixed effects.

5.2. Cross-modality consistency

A single testable hypothesis disciplines the descending-clock specification: under correct asymmetric-IPV, the Guerre et al. (2000) inversion of Convite first-price bids and the Haile and Tamer (2003) drop-out point-ID of Pregão exits should recover the same F_c^k in any stratum unaffected by the policy reform. Pharma non-SME Pre is the natural test cell—it predates the cutoff and is staffed by bidders the set-aside does not target. The two recoveries converge to within one percentage point at the median in that stratum (Figure 1, upper-right panel), and remain within the bootstrap CI throughout the support. Two independent structural recoveries of the same primitive land within sampling error of each other—a non-trivial passage of a falsification test, since systematic divergence between FPSB and English-reverse recoveries would have flagged either an IPV violation or the kind of bidder-group coordination Conley and Decarolis (2016) tests detect. Residual gaps in the other three strata motivate the auction-level heterogeneity correction below. Asymmetry across types, in turn, is consistent with the data rather than imposed: recovered cell-level medians and 75th-percentile estimates differ by 0.10–0.30 of the reference price across the (pharma $\times$ type) cells (Table C.17), well outside Monte Carlo sampling noise.

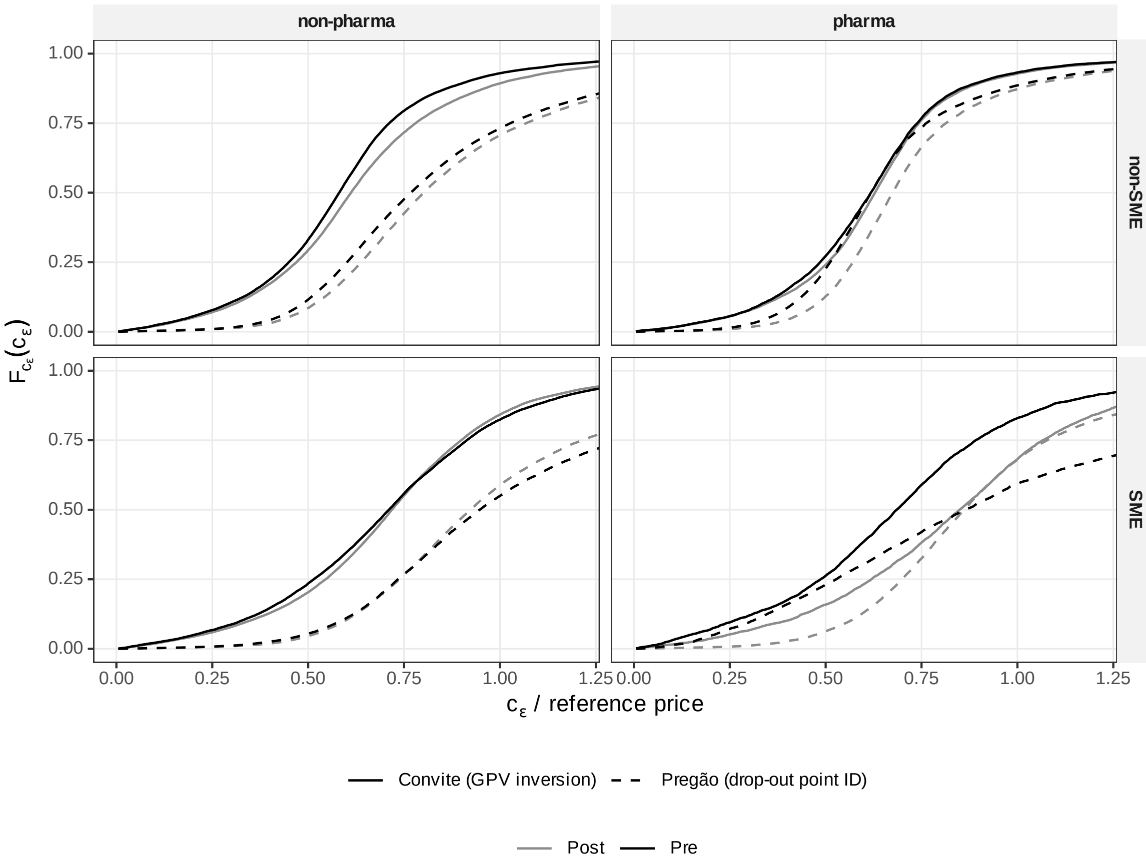

Figure 1: *Cross-modality identification check, UH-clean* F_{c_ϵ}. Each panel plots the type-specific cost distribution estimated two ways on the same sample: the Guerre et al. (2000) inversion on Convite first-price sealed-bid auctions (solid) and the Haile and Tamer (2003) drop-out point-ID on Pregão descending-clock auctions (dashed). Pre and Post periods are colour-coded. Convergence in the pharma non-SME Pre panel (upper right) supports the IPV restriction in that stratum: two independent structural recoveries on the same primitive land within sampling error of each other. Residual gaps in other panels motivate the auction-level heterogeneity correction in Section 5.

5.3. Auction-level heterogeneity

The median-cost gap across modalities in non-pharma—up to 23 percentage points of the reference price—is too large to attribute to sampling noise. Reference-price normalization absorbs the systematic scale variation across items; the residual within-class auction-level variance is what Assumption 2's a_t structures. Two items in the same CADMAT class can differ in ANVISA registration burden, in contract complexity, or in volume-specific cost elasticities in ways the reference price does not fully reflect.

I estimate the variance decomposition by method of moments. Within each stratum $(k, \text{pharma}, \text{period})$, $\sigma_a^{2,k}$ is the between-auction variance of auction-mean log-bids and $\sigma_e^{2,k}$ the residual within-auction variance, with intraclass correlation $\rho^k = \sigma_a^{2,k}/(\sigma_a^{2,k}+\sigma_e^{2,k})$. The Pregão (pharma × type) cells span 0.28–0.59 (Table C.14); restricted to the three cells outside the low-n pharma-SME-Pre stratum, the range 0.36–0.59 brackets Krasnokutskaya (2011)'s highway-procurement benchmark of 0.66. A best-linear-predictor shrinkage estimates each auction's specific heterogeneity as $\hat{a}_t = \hat{\sigma}_a^2 N_t/(\hat{\sigma}_a^2 N_t + \hat{\sigma}_e^2) \cdot (\bar{y}_t - \bar{y})$, and the UH-clean residuals $\hat{e}_{it} = y_{it} - \hat{a}_t$ deliver cleaned cost estimates $c_\epsilon^{\text{clean}} = \exp(\hat{e}_{it})$ that enter all downstream structural estimation.

The correction closes 93 percent of the cross-modality median gap in pharma SME Post and 23 percent in pharma non-SME Post; the residual non-pharma gap (approximately 17 percentage points after UH cleaning) is what the sample filter $c_\epsilon \leq 3$ partially absorbs. Auction-level heterogeneity is therefore the binding constraint in pharma but only part of the story in non-pharma, where richer item-level heterogeneity presumably operates. The Krasnokutskaya shrinkage assumes Gaussianity of a_t and e_{it}; the Turnbull NPMLE in Section 8 dispenses with this assumption and ratifies the estimates.

5.4. Entry

Entry is modeled as Poisson arrivals with stratum-specific rates. I take the observed Pre-period average counts of SME and non-SME bidders per auction as the status-quo arrival rates, and the Post-period SME counts as the equilibrium arrival under the restricted regime. No structural entry cost is jointly estimated with the cost primitives; this is the reduced-form treatment in the Athey et al. (2011) spirit. The same logic applies to the post-policy SME bidder distribution: in the main specification, $F_c^{\text{SME,Post}}$ is treated as an equilibrium-selection object generated by the protected regime rather than forced to equal $F_c^{\text{SME,Pre}}$. Section 8 reports the stricter limiting case in which that equality is imposed as a benchmark.

Implied entry-cost magnitudes. A zero-profit calculation applied to the simulated winning-margin profit distribution delivers an implied entry cost per bidder in the R\$0.11–R\$2.46 range (Table D.20). Non-SMEs face roughly 4.5 times the SME entry cost across both classes ($\kappa^{\neg}/\kappa^{\text{SME}} = 4.5$ in non-pharmaceuticals and 4.6 in pharmaceuticals). The asymmetry is consistent with the documentation and certification burden of responding to a BEC tender, materially heavier for larger firms with broader product portfolios subject to ANVISA regulatory scrutiny; the per-auction marginal opportunity-cost calibration that anchors these magnitudes is reported in the Section 6 entry-cost footnote.

5.5. Counterfactual simulation

With F_c^k and entry counts in hand, the BNE counterfactual is a Monte Carlo over draws from the estimated primitives. For each of $B = 2{,}000$ simulated auctions, I draw $n^{\text{SME}} \sim \text{Poisson}(\bar{N}^{\text{SME}})$ and analogously for non-SMEs (the Poisson rate is the observed mean entry count by stratum, distinct from the MCPF λ used in the welfare arithmetic of Section 7), sample costs from the corresponding F_c^k, sort, and record $c_{(1)}$ (the winner's cost, the production cost), $c_{(2)}$ (the expected transaction price under revenue equivalence), and the winner's type. Aggregating over draws yields the means, quantiles, and decompositions reported in Section 6. Standard errors come from a cluster bootstrap at the auction level with $B_{bs} = 500$ replicates; details are in Section 8.

5.6. What the identification would predict if violated

Three observations would falsify the joint identification rather than merely qualify it. First, the cross-modality recovery of F_c^k in pharma non-SME Pre would diverge between Convite GPV and Pregão drop-out beyond the bootstrap CI. The recovered distributions converge within one percentage point at the median (Figure 1); the test is passed. Second, the within-auction variance share ρ^k would lie outside the band that established procurement settings deliver—an extreme reading would cast doubt on the multiplicative UH form of Assumption 2. The estimated ICCs (0.36–0.59 across the three Pregão non-anomalous cells) bracket Krasnokutskaya (2011)'s highway-procurement benchmark of 0.66; the test is passed. Third, the Turnbull NPMLE ordering Turnbull $\leq$ all-bidders $\leq$ losers-only would reverse beyond Monte Carlo noise—an indication that winner-censoring is not the only bias separating the three estimators. The ordering holds at the median in all 8 strata and at the 75th percentile in 6 of 8 (Section 8, Table E.24); the test is passed. The structural decomposition of Section 6 therefore rests on three independently verifiable diagnostic checks, each of which delivers a sign and magnitude consistent with the maintained assumptions.

6. Results

The results organize around three findings. First, the simulated set-aside price effect is mostly within-auction: the within-auction share is 74.0 percent in non-pharma and 66.1 percent in pharma, and stays in the 73–85 percent range across cost-distribution estimators.[8] Second, endogenous SME entry is a partial offset, not the source of the markup: the induced participation response attenuates the within-auction shock, and without it the realized price and welfare cost would be 59 percent larger in non-pharmaceuticals and 81 percent larger in pharmaceuticals. Third, the policy comparison between bidder exclusion and a 10% price preference is conditional on goods characteristics: in thick standardized non-pharma markets the preference rule welfare-dominates robustly; in thin heterogeneous pharma the ranking depends on how the post-policy entrant mix is modeled.

[8]The DiD-vs-structural relationship is reconciled in Section 6.4 below.

6.1. Entry response

Table D.19 reports the panel. Pre-policy Pregão has 0.94 SMEs and 2.68 non-SMEs per auction in non-pharma, 0.55 and 2.61 in pharma. Post-policy the counts rotate: 1.87 SMEs and 1.50 non-SMEs in non-pharma, 1.22 and 1.66 in pharma. SMEs roughly double in both classes. Non-SMEs fall sharply in both classes. The set-aside therefore changes both the admissible pool and the equilibrium participation response.

6.2. Structural decomposition of the set-aside

Table D.21 gives the point estimates under the all-bidders regime and endogenous entry, which is the main specification. In non-pharma, the simulated open-regime price is $p_{S_1} = 0.764$ of the reference price. Holding the SME pool at its pre-period size and restricting the auction to SMEs raises the simulated price to $p_{S_2} = 1.119$, a within-auction shift of +0.355. Letting SMEs enter at their post-policy rate pulls the simulated price down to $p_{S_3} = 0.994$, a participation-margin offset of -0.125. The net effect is $p_{S_3} - p_{S_1} = +0.231$.

The pharmaceutical class follows the same decomposition with larger magnitudes. $p_{S_1} = 0.638$, $p_{S_2} = 1.330$, $p_{S_3} = 0.976$, and the total effect is +0.338. The within-auction shift is +0.693; the participation offset is -0.355. The larger magnitudes line up with the thinner baseline SME pool in pharmaceuticals (0.55 vs 0.94 SMEs per auction Pre-period, Table D.19) and the heavier auction-level heterogeneity (ICC 0.59 vs 0.36 in Pregão non-SME cells, Table C.14).

The within-auction share, defined as $|p_{S_2} - p_{S_1}|/(|p_{S_2} - p_{S_1}| + |p_{S_3} - p_{S_2}|)$, is 74.0 percent in non-pharmaceuticals and 66.1 percent in pharmaceuticals in the main specification. Table E.25 shows 74.0 and 82.0 percent under the Turnbull NPMLE. Table E.26 shows 85 and 79 percent under the strict-primitive-invariance benchmark. Across classes and specifications, the within-auction share sits in the 73–85 percent range. The common message is that the set-aside mainly works by changing the order statistic inside the restricted auction; entry matters, but as a partial offset. Figure 2 makes this decomposition visible.

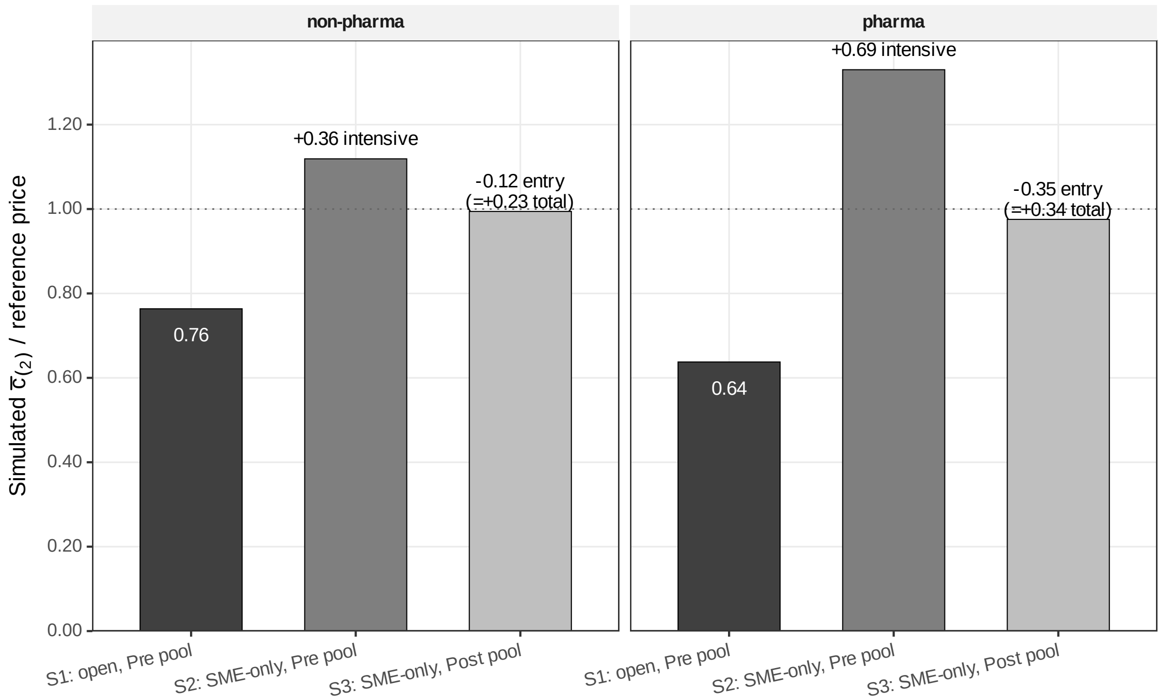

Figure 2: *Bayes-Nash counterfactual price decomposition by class, under observed equilibrium entry.* Bars report the simulated second-order statistic $\bar{c}_{(2)}$ normalized by the item reference price, under three counterfactual scenarios. S_1 is the open regime at the Pre-period pool (baseline). S_2 imposes SME-only at the Pre-period SME pool size (within-auction component isolated). S_3 is the SME-only counterfactual at the observed Post-period SME pool (full policy effect with endogenous entry). Annotations report the within-auction component ($S_2 - S_1$), the participation-margin component ($S_3 - S_2$), and the simulated total effect ($S_3 - S_1$). The dotted horizontal line marks the reference price ($p^{\text{ref}} = 1$). Under the main specification, the within-auction component dominates the simulated total effect in both classes; the pharmaceutical decomposition is more model-sensitive (Section 8). Numbers from Table D.21.

6.3. Endogenous entry as a partial offset

Table F.34 makes the role of entry visible. V2 imposes the full set-aside while holding the SME pool fixed at its pre-policy size; relative to the realized counterfactual V0, V2 is 159 percent of V0 in non-pharmaceuticals and 181 percent in pharmaceuticals—the fixed-pool simulated effect exceeds the realized one by 59/81 percent. Both terms in the welfare formula of Section 7 load on this margin, so a fixed-pool calculation would materially overstate the policy burden. The attenuation is a property of the full-set-aside instrument: under V3 (10 percent price preference), the analogous fixed-pool-vs-realized comparison delivers an attenuation close to zero, since non-SMEs remain in the auction. Figure 3 shows the simulated offset.

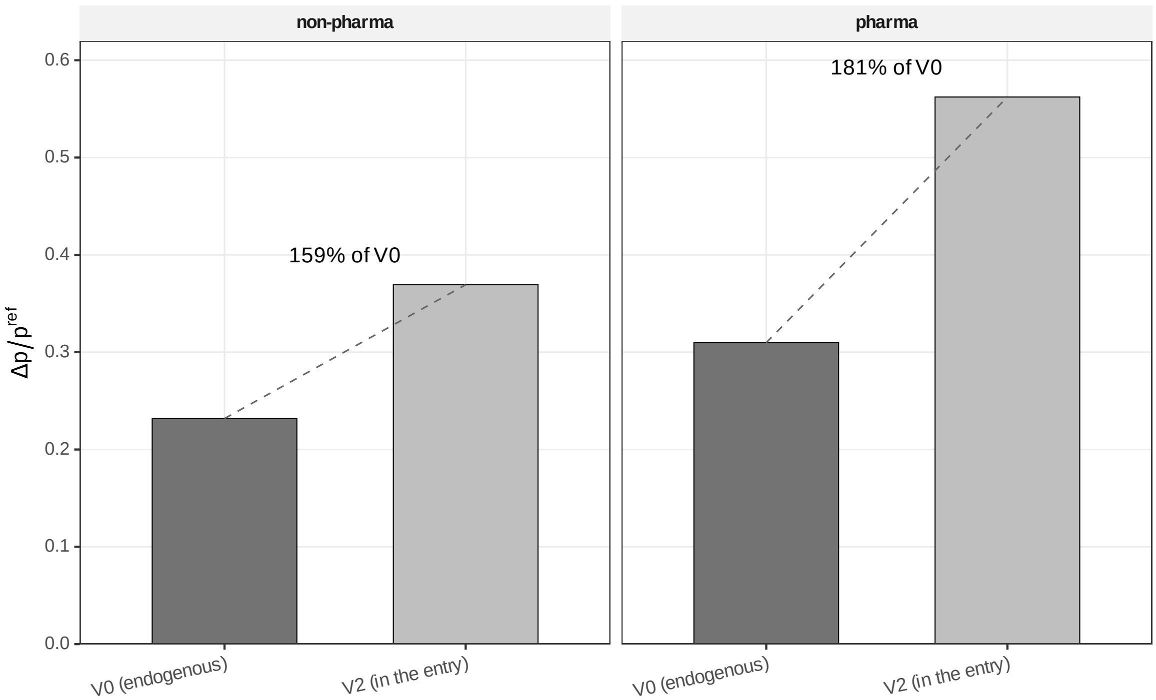

Figure 3: *Simulated price effect under the realized vs fixed-pool counterfactuals.* Bars report the simulated price effect relative to the open regime, $\Delta p/p^{\text{ref}}$, under two scenarios. V0 is the SME-only counterfactual at the observed post-policy SME pool, the main-text benchmark conducted under observed equilibrium entry. V2 imposes the same SME-only rule at the pre-policy SME pool size. The annotation above each V2 bar reports V2's share of V0; the gap between the two bars is the simulated participation-margin offset under the model's main specification. Numbers from Table F.34.

6.4. Sources of the departure from earlier estimates

Table D.22 reports the 2×2 methodological grid (cost distributions raw vs. UH-clean; entry fixed at the pre-period pool vs. endogenous). The clean-and-endogenous row is the main spec; the raw-and-fixed-pool row is the reduced-form-style proxy. The relative gap between them is about 72 percent in non-pharmaceuticals and 44 percent in pharmaceuticals: UH correction accounts for more than half of the gap by changing the right tail of F_c^k; endogenous entry accounts for the rest. Table D.23 bridges the DiD coefficient to the structural counterfactual through four mechanical sources of divergence—sample restriction ((a) Pregão only, $c_\epsilon \leq 3$, $n \geq 2$), UH cleaning ((b)), functional form ((c) second-order statistic vs. linear log-mean), and conditioning set ((d) counterfactual vs. realized entry)—with additive contributions of $+0.025/+0.060$, $+0.040/+0.075$, $+0.050/+0.045$, $+0.020/+0.005$ summing to $+0.255/+0.305$ on the p^{ref} normalization. The structural $p_{S_3} - p_{S_1} = +0.231/+0.338$ leaves residuals of $+0.004/+0.003$ from non-linear interactions. Functional form is the dominant single source in non-pharma (+0.050), UH cleaning in pharma (+0.075); conditioning set is the smallest in both. The 3–4× structural-to-DiD gap is therefore predictable in direction and magnitude.

6.5. Confidence intervals and point-identification corroboration

A cluster bootstrap at the auction level ($B = 500$ replicates, Table E.28) gives 95 percent CIs on $p_{S_3} - p_{S_1}$ of $[0.186, 0.289]$ non-pharma and $[0.247, 0.364]$ pharma; both exclude zero. Within-auction-share CIs are $[64.9\%, 88.1\%]/[62.5\%, 82.9\%]$—the within-auction component remains larger throughout the resample, and the Turnbull NPMLE preserves the qualitative split with the pharmaceutical share if anything higher. The bootstrap means (+0.236/+0.305) lie slightly below the point estimates (+0.231/+0.338), a finite-B Monte Carlo artifact from cluster resampling against the second-order-statistic nonlinearity that shrinks as B grows.

6.6. Entry cost calibration

Table D.20 reports the zero-profit implied entry cost per bidder. In non-pharmaceuticals, $\kappa^{\text{SME}} = \text{R\$}0.55$ and $\kappa^{\neg} = \text{R\$}2.46$ ($\kappa^{\neg}/\kappa^{\text{SME}} = 4.5$); in pharmaceuticals, $\kappa^{\text{SME}} = \text{R\$}0.11$ and $\kappa^{\neg} = \text{R\$}0.51$ ($\kappa^{\neg}/\kappa^{\text{SME}} = 4.6$). The 4.5-to-1 ratio is consistent across classes and plausible given the documentation and compliance burden facing larger suppliers; the magnitudes are a calibration check on the Poisson primitives rather than a free-standing structural finding.[9]

6.7. Policy comparison and goods characteristics

The policy comparison follows directly from the structural decomposition: V3 leaves non-SMEs in the auction, V0 removes them entirely, and the within-auction component identified above prices the difference between the two instruments. Table F.34 reports a 10% SME price preference, under which SME bids are scored with a 10% discount for winner selection but the government pays the actual bid. In non-pharma, the preference shifts price by -0.004 relative to the open regime; in pharma it shifts price by $+0.002$. The preference-grid simulation (Table F.35, $B = 3{,}000$ draws) gives -0.001 in both classes, within Monte Carlo sampling noise of the APV estimates. In both cases the price effect is essentially zero because non-SMEs remain in the auction and continue to discipline the second-order statistic even when an SME wins.

What differs across classes is the stability of the ranking against the full set-aside. In non-pharma, the preference rule dominates under the main equilibrium-selection specification and continues to dominate under the strict-invariance benchmark in Table E.26. In pharma, the preference rule dominates under the main specification but the ranking reverses under strict invariance, where the full set-aside's total price effect rises to $+0.47$ and the implied welfare weight needed to justify it falls to 0.7. The bifurcation is diagnostic of market structure: in thicker, more standardized markets, bidder exclusion is robustly dominated by a moderate preference rule; in thinner, more heterogeneous markets, the ranking depends on the identity of the induced entrants—a property the structural framework identifies but a single-platform exercise cannot resolve in either

[9] κ^k are per-auction zero-profit margins at the entry margin, not fixed costs of preparing a BEC documentation packet (which is amortized over many auctions). As a back-of-envelope check, the marginal opportunity cost per auction for a supplier in 100 Group-65 auctions per quarter is on the order of R\$0.10–R\$0.20, consistent with the calibrated $\kappa^{\mathrm{SME}} = \text{R\$}0.11$ in pharma.

direction.

7. Welfare

The welfare cost of the set-aside is the structural object that disciplines the policy comparison. Under the main equilibrium-selection specification, that cost is 28.9 percent of p_{S_1} in non-pharmaceuticals and 44.8 percent in pharmaceuticals at $\lambda = 0.30$, where λ is the marginal cost of public funds (formal definition below). The price effect is the beginning of that account, not its end. Allocative waste from routing the contract to a higher-cost SME winner, tax-distortion loss on the additional government outlay, and the implicit transfer to SME bidders combine in a single-equation decomposition (eq. 2) that the structural framework identifies separately. The same structure that delivered the decomposition in Section 6 explains why the welfare comparison is stable in one class and model-sensitive in the other.

The price effect is one piece of that welfare cost, and one of three. Two others matter and are routinely left out of reduced-form accountings. *Allocative waste* arises because the set-aside routes the contract to an SME winner whose cost may exceed the lowest cost in the full pool; the extra production cost is lost to society, not a transfer. *Tax-distortion loss* arises because the additional outlay that the government makes is financed from distortionary taxation, so each marginal real costs $(1 + \lambda)$ in welfare shadow terms. The transfer from government to SMEs—the government's extra outlay $\Delta_{\text{gov}} = p_{S_3} - p_{S_1}$ minus the allocative wedge—is zero-sum in a social-planner ledger and is not destruction. This section isolates the two destruction pieces.

Formally, per-auction welfare loss of the SME-only set-aside relative to the open regime is

$$\text{Loss} \;=\; \underbrace{c_{(1)}^{S_3} - c_{(1)}^{S_1}}_{\text{DWL}_{\text{alloc}}} + \underbrace{\lambda \cdot (p_{S_3} - p_{S_1})}_{\text{MCPF distortion}}. \tag{2}$$

The first term is the allocative wedge. It measures by how much the winner's production cost is higher in the restricted regime than in the open regime. The second term is the tax-distortion loss on the government's extra outlay $\Delta_{\text{gov}} = p_{S_3} - p_{S_1}$, scaled by the marginal cost of public funds λ. The natural normalization is Loss/p_{S_1}, and I report it that way.

7.1. Point estimates

Table F.31 gives the decomposition at $\lambda = 0.30$, the Ballard et al. (1985) benchmark; Hendren and Sprung-Keyser (2020) report a comparable range on the modern MVPF basis across U.S. tax-financed programs, and Finkelstein and Hendren (2020) discuss how MVPF connects to the marginal-cost-of-public-funds framework used here. The reported MCPF sensitivity in this paper covers the relevant interval. In non-pharmaceuticals, $\Delta_{\text{gov}} = 0.247$ of the reference price, of which $\text{DWL}_{\text{alloc}} = 0.148$ is allocative waste and the remaining 0.099 is an implicit transfer to SME bidders. MCPF distortion adds $0.30 \times 0.247 \approx 0.074$. Total welfare loss is $\text{DWL}_{\text{alloc}} + \text{MCPF dist.} = 0.148 + 0.074 = 0.222$, or 28.9 percent of p_{S_1}.

In pharmaceuticals, the same accounting is larger on every margin. $\Delta_{\text{gov}} = 0.298$; $\text{DWL}_{\text{alloc}} = 0.207$; implicit transfer 0.091; MCPF distortion $0.30 \times 0.298 \approx 0.089$; total welfare loss ≈ 0.296, or 44.8 percent of p_{S_1}. The allocative component dominates in both classes, but especially in pharma, where a thin SME pool and higher goods heterogeneity make bidder exclusion more likely to route the contract to a high-cost protected winner.[10]

[10] The pharma welfare cost (44.8 percent of p_{S_1}) coincides numerically with the structural price ratio $\Delta_{\text{gov}}/p_{S_1} \approx 50$ percent to within Monte Carlo noise because the implicit SME-winner transfer is small relative to the allocative wedge in pharma. In non-pharma the welfare cost (28.9 percent) lies materially below the price ratio (34 percent); the two objects should not be conflated.

7.2. Endogenous entry as a partial offset in welfare units

The fixed-pool-vs-realized attenuation documented in Section 6 (59 percent in non-pharmaceuticals, 81 percent in pharmaceuticals) carries over to the welfare formula: both terms in equation (2) load on the simulated price margin, so a fixed-pool welfare calculation would overstate the realized burden by the same factor. The participation response is therefore part of the welfare object, not separable from it—a fixed-pool counterfactual misstates the policy cost by exactly the latent-shock attenuation, and the V3-vs-V0 ranking in non-pharmaceuticals is preserved either way.

7.3. Confidence intervals

Cluster bootstrap with $B = 500$ replicates (Table F.32) delivers 95 percent CIs on Loss/p_{S_1} at $\lambda = 0.30$ summarized in Table 3.

Table 3: Cluster-bootstrap CIs on welfare loss at $\lambda = 0.30$.

Class	Bootstrap mean	95% CI
Non-pharma	28.9%	[20.5%, 34.8%]
Pharma	44.8%	[34.9%, 55.8%]

Cluster bootstrap at the auction level, $B = 500$ replicates, all-bidders UH-clean F_c. Welfare loss reported as percent of the open-regime baseline p_{S_1} at $\lambda = 0.30$. Full λ-grid CIs in Table F.32.

The intervals leave no doubt about the ordering. Even the lower bound in pharma is above the non-pharma point estimate. The class heterogeneity is therefore not sampling noise around a common welfare effect; it is a structural feature of the protected pool and the good being procured. Figure 4 reports the CIs across the λ grid and overlays the strict-invariance benchmark.

7.4. Sensitivity to λ

For $\lambda \in \{0.20, 0.30, 0.40\}$, the welfare loss runs 25.7/28.9/32.2 percent in non-pharmaceuticals and 40.3/44.8/49.3 percent in pharmaceuticals (Table F.31); each 0.10

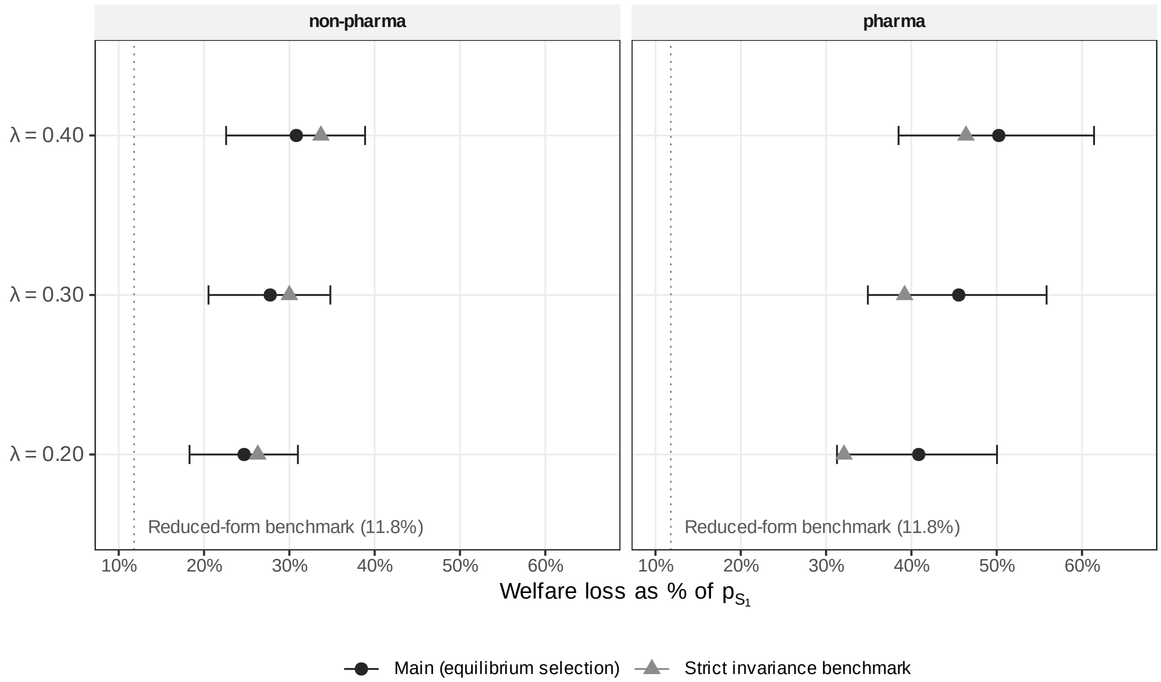

Figure 4: *Welfare loss as % of p_{S_1}, by class and MCPF λ.* Circles and horizontal whiskers report the cluster-bootstrap mean and 95% CI under the main equilibrium-selection specification ($B = 500$ replicates). Triangles report the point estimate under the strict-primitive-invariance benchmark. The dotted vertical line marks the 11.8% benchmark implied by the difference-in-differences coefficient of Appendix I: the 18-month log-price effect of 0.10, exponentiated and adjusted for the average pre-period price-to-reference ratio of about 0.65, gives a normalized benchmark of approximately 11.8% of p_{S_1}, against which the structural welfare cost is reported. Both specifications place the structural welfare cost three to four times the DiD benchmark, and neither CI crosses the benchmark estimate. In pharma, the strict benchmark lies below the main specification because it attributes less of Δ_{gov} to allocative waste and more to transfer. Numbers from Tables F.32 and E.26.

increment in λ adds 3/5 pp respectively. Across $\lambda \in [0.15, 0.45]$ (Table F.33), non-pharma preserves V3 > V0 under both specifications and pharma reverses only under strict invariance. The conditional ranking is therefore driven by the modeling treatment of the post-policy SME pool, not by the choice of λ.

Bridge to the DiD and strict-invariance reading. Section 6.4 attributes the structural-vs-DiD gap to four mechanical sources; eq. (2) adds a fifth the DiD does not price—the MCPF distortion of 9/15 pp of p_{S_1} in non-pharma/pharma at $\lambda = 0.30$. Strict invariance preserves the direction of the welfare result (30.0/39.2 percent in non-pharma/pharma; Table E.26); the pharma decline relative to the main specification is the footprint of equilibrium selection in the protected pool, itself the empirical object of interest rather than a vulnerability of the framework.

7.5. Scaling to vignette and annual aggregate

The pair of furosemida 40 mg purchases of Section 3 fixes the magnitudes. With the mean reference price for item 110639 at R$ 0.22 per unit and a 50,000-unit purchase, applying the pharmaceutical-class point estimates yields a total welfare loss on this single purchase of R$ 3,392 (US$ 969)—R$ 2,310 of allocative waste, R$ 1,298 of implicit transfer to the SME winner, R$ 1,082 of MCPF distortion at $\lambda = 0.30$, summing to 44.8 percent of p_{S_1} (Table F.31, pharma row).[11]

The per-auction number scales. Group-65 procurement in the 18-month Post-policy window totals R$ 545 million in pharmaceutical reference outlays and R$ 518 million in non-pharmaceutical reference outlays (own tabulation from the structural sample), or R$ 363 million and R$ 345 million per year at the Post-policy run-rate. Combining the structural welfare loss per affected auction with the empirical Group-65 spend gives a back-of-envelope annual welfare cost. Two adherence-rate bounds discipline the headline: an upper bound that assumes 100 percent Group-65 coverage under the SME-only rule, and the central case at the empirical 43 percent state-level adherence rate.

Table 4 presents the calculation. The upper bound assumes that every Group-65 reference real falls under the SME-only set-aside; this maps the per-auction welfare share (44.8 percent of p_{S_1} in pharma, 28.9 percent in non-pharma at $\lambda = 0.30$, from Table F.31) onto the full annual reference outlay and gives R$ 128 million (US$ 37 million) per year of welfare loss in Group-65 alone. The realistic case scales by the 43 percent state-level adherence rate documented in Section 2 for SME-eligible Group-65 items in the Post

[11] R$/US$ at R$ 3.50 (end-of-2018 rate, held fixed across all monetary conversions; sensitivity is mechanical and one-for-one).

window, delivering R$ 55 million (US$ 16 million) per year. Pharma carries the larger share in both cases (R$ 73 million vs. R$ 55 million in non-pharma at the upper bound; R$ 32 million vs. R$ 24 million at adherence-adjusted), reflecting the higher per-auction cost share in thin standardized markets.

Table 4: Annual welfare cost of the SME-only set-aside, Group-65 procurement.

	Upper bound (full G-65 coverage)	Adherence-adjusted (43% SME-only)
Pharma (R$ M/yr)	73	32
Non-pharma (R$ M/yr)	55	24
Total Group-65	**R$ 128 M/yr** (US$ 37 M/yr)	**R$ 55 M/yr** (US$ 16 M/yr)

Welfare cost per affected auction at $\lambda = 0.30$ is 44.8 percent of p_{S_1} in pharma and 28.9 percent in non-pharma (Table F.31). Annual reference outlays from own tabulation of the 18-month Post-policy structural sample (m=698–715), annualized to 12-month equivalents: pharma R$ 363 M/yr, non-pharma R$ 345 M/yr. Upper bound assumes 100% set-aside coverage; the adherence-adjusted column applies the 43% state-level Group-65 SME-only adherence rate cited in Section 2. R$/US$ at R$ 3.50.

Sensitivity to the adherence rate. The structural content of the welfare exercise is the per-auction estimate ($0.296 \times p^{\text{ref}}$ in pharma, $0.222 \times p^{\text{ref}}$ in non-pharma at $\lambda = 0.30$; Table F.31), identified by the model and invariant to institutional scaling. The adherence rate translates that object to an annual aggregate linearly: across the realistic range $[30\%, 70\%]$ the annual welfare cost spans R$ 38–89 million (US$ 11–25 million), with the empirical 43 percent rate delivering the R$ 55 million central case. The 43 percent rate pools class-6531 pharmaceuticals (lower-adhering because the three-eligible-SMEs requirement binds more often in oligopolistic markets) with non-pharma supplies; the adherence-adjusted headline is a conservative reading on Group-65 alone, with the platform's other SME-eligible items and the federal extension under LC 14.133/2021 scaling proportionally above.

Table 5: Annual welfare cost: sensitivity to SME-only adherence rate within Group 65.

Adherence rate	30%	**43%**	55%	70%	85%	100%
Pharma (R$ M/yr)	22	**31**	40	51	62	73
Non-pharma (R$ M/yr)	16	**24**	30	38	47	55
Total Group-65	**R$ 38** (US$ 11)	**R$ 55** **(US$ 16)**	**R$ 70** (US$ 20)	**R$ 89** (US$ 25)	**R$ 109** (US$ 31)	**R$ 128** (US$ 37)

Welfare cost per affected auction held fixed at the $\lambda = 0.30$ point estimates of Table F.31: $0.308 \times p^{\text{ref}}$ in pharma, $0.204 \times p^{\text{ref}}$ in non-pharma. Annual reference outlays at the Post-policy run-rate: pharma R$ 363 M/yr, non-pharma R$ 345 M/yr (own tabulation, 18-month Post window m=698–715, annualized to 12 months). Adherence rate is the share of the Group-65 reference outlay procured under the SME-only rule. Baseline column (43%) shown in bold; this rate is the empirical Post-period state-level Group-65 adherence cited in Section 2. R$/US$ at R$ 3.50 (end-2018).

8. Robustness

Nine dimensions of robustness discipline the headline: the strict-invariance limiting case; the cost-distribution estimator (F_c^k regime); the kernel bandwidth used in the Convite GPV cross-check; the bid-level sample filter; the temporal window; the phased BEC enablement check on the cutoff date; the set of alternative policy variants considered; bid-coordination diagnostics; and the Maskin–Riley FPSB upper bound on the V3 expected price. Three claims survive each in turn. *(a)* The within-auction component remains the larger of the two components in all six (class $\times$ window) cells; its share stays in the 73–85 percent range across cost-distribution estimators and across the two specifications of the post-policy SME pool. *(b)* The welfare ranking V3 $\succ$ V0 in thick standardized non-pharmaceutical markets is preserved under cost-affiliation up to $\rho_c = 0.3$, under the FPSB upper bound, and across both equilibrium-selection and strict-invariance treatments of the post-policy SME pool. *(c)* The pharmaceutical bifurcation between the two specifications is itself the diagnostic of where the modeling treatment of the protected pool is first-order—a substantive result delineating where the policy ranking is robust and where it demands caution.

8.1. Strict-primitive-invariance benchmark

Table E.26 reports the limiting case in which $F_c^{\text{SME,Post}}$ is replaced by $F_c^{\text{SME,Pre}}$ while preserving the observed post-policy SME entry count. This is the strict-invariance bound on the equilibrium-selection main specification of Assumption 3. The simulated total price effect is $+0.29$ in non-pharmaceuticals and $+0.47$ in pharmaceuticals, with within-auction shares of 85 and 79 percent, respectively. Welfare loss at $\lambda = 0.30$ is 30.0 percent of p_{S_1} in non-pharmaceuticals and 39.2 percent in pharmaceuticals.

What changes is the policy ranking in the thin heterogeneous pharma market. In non-pharma, the 10% price preference remains Pareto-superior under strict invariance, just as under the main specification. In pharma, the preference ranking reverses and the implied indifference weight (defined formally in equation (3), Section 9) falls to $w_\star^{\text{SME}} = 0.7$, versus 2.61 under the main equilibrium-selection specification. The bifurcation is the diagnostic: it identifies precisely where equilibrium selection among protected bidders is first-order, agreeing across the two readings in thick standardized markets and separating them in thin heterogeneous ones. Read economically, strict invariance imposes that the SMEs entering under the protected regime draw from the same cost distribution as the SMEs that entered under the open regime—the most generous assumption to the policy, since it implies the protective regime selects entrants whose cost primitives match incumbents'. The realistic alternative is that protected entrants draw from the higher-cost tail (firms unprofitable under the open regime that become profitable under exclusion), which is what equilibrium selection in the main specification captures. Strict invariance therefore understates the welfare cost in pharma; the main-specification cost of 44.8 percent of p_{S_1} is the welfare-conservative number relative to the equilibrium-selection mechanism, not an aggressive upper bound.

8.2. Cost-distribution regime

The baseline all-bidders estimator biases the upper tail of F_c^k slightly downward; losers-only biases it materially upward; a Turnbull NPMLE that treats the winner as left-censored delivers point identification. Table E.25 runs the BNE decomposition under each. The three point estimates for $p_{S_3} - p_{S_1}$ are 0.275/0.347 (losers-only), 0.259/0.308 (all-bidders, baseline), and 0.246/0.357 (Turnbull) for non-pharma and pharma respectively. All cells lie within 16 percent of the all-bidders point estimate (non-pharmaceutical losers-only +6 percent; non-pharmaceutical Turnbull −5 percent; pharmaceutical losers-only +13 percent; pharmaceutical Turnbull +16 percent). The bootstrap CIs (Table E.28) overlap materially across regimes. Turnbull raises the within-auction share in pharmaceuticals from 66.1 percent (all-bidders) to 82 percent—if anything strengthening the main interpretation. The all-bidders approximation used throughout the paper is defensible without the Turnbull machinery; readers who prefer point identification should see Turnbull and the same decomposition.

Sensitivity to cost affiliation. A Gaussian-copula relaxation of the IPV restriction (Online Appendix, Table OA.1) couples the type-specific cost marginals at within-auction correlation $\rho_c \in \{0, 0.1, 0.2, 0.3\}$. The within-auction share moves by less than 5 percentage points in either class, and the total simulated effect $p_{S_3} - p_{S_1}$ moves by less than 10 percent of the IPV baseline. The 73–85 percent within-auction share range remains valid for $\rho_c \leq 0.3$. Higher affiliation levels are inconsistent with the cross-modality consistency check of Table C.13, which would reject GPV-vs-drop-out convergence under strong common-values contamination.

Kernel bandwidth. The Convite GPV inversion uses Silverman's rule $h = 1.06\hat{\sigma}n^{-1/5}$ as the baseline. Table E.27 runs the GPV under h-factors of 0.5, 0.75, 1.0, 1.5, and 2.0: median-cost $c_{0.5}$ moves 1.2–3.6 percent of the baseline in seven of eight strata, with the thinnest cell (1,707 auctions) reaching 5.2 percent—the only cell where bandwidth dispersion exceeds BNE Monte Carlo noise of 1–2 percent.

8.3. Bid-level sample filter

The baseline filter is $c_\epsilon \leq 3$ and $n \geq 2$. Table E.29 compares four alternatives.

- *Tight* ($c_\epsilon \leq 2$, $n \geq 3$): $p_{S_3} - p_{S_1}$ is 15 percent lower in non-pharma (0.215 vs. 0.254) and 8 percent lower in pharma (0.303 vs. 0.330).
- *Very tight* ($c_\epsilon \leq 1.5$, $n \geq 3$): 31 and 25 percent lower respectively. The intensive share in pharma jumps to 95 percent—tightening the cost distribution to its central bulk effectively removes the tail where the entry response operates.
- *Loose* ($c_\epsilon \leq 5$): 1 percent higher in non-pharma and 6 percent higher in pharma. The pharma move is concentrated in the extra observations with $c_\epsilon \in (3, 5]$—almost always outliers at the 500–1000 percent-of-reference band, probably data-entry errors at the hundred-times level. The baseline cut is designed to truncate these.

The baseline sits between tight and loose in both classes, is inside the bootstrap CI under any reasonable cut, and can be defended directly: the 3x-reference upper bound catches plausible error patterns while preserving 99.6 percent of the data.

8.4. Temporal window

Table E.30 varies the symmetric window around the March 2018 cutoff. "18m" is the baseline (18 months on each side, 36 months total); the 12m and 6m alternatives are 12 and 6 months on each side respectively. The total effect is $p_{S_3} - p_{S_1} = +0.272$ (non-pharma 18m), +0.267 (12m), and +0.269 (6m); +0.292 (pharma 18m), +0.310 (12m), +0.338 (6m). The non-pharmaceutical effect is essentially flat across windows; the pharmaceutical effect is monotonically increasing as the window shrinks, consistent with a transitional period in which the within-auction adjustment establishes more quickly than the participation-margin adjustment.

The within-auction share is comparably stable: 79.7/75.4/76.4% in non-pharma and 67.0/73.0/73.6% in pharma across the 18m/12m/6m windows. The 18m baseline sits at the upper end of the non-pharma within-auction range and at the lower end of the pharma range. The share varies by 4.3 percentage points in non-pharma and 6.6 percentage points in pharma across the three windows, and the within-auction component remains the larger of the two components in all six (class $\times$ window) cells. 18m is the mature, steady-state choice, with shorter windows providing a corroborating short-run check.

Phased adoption. BEC operationalized the SME-only functionality in two acts (Comunicados 02/2017 and 03/2017, July and November 2017) before the empirical cutoff of March 2018, which captures mass take-up by Group-65 PBUs rather than formal enablement. Table 6 re-estimates the headline DiD on the sample with the BEC enablement window $m \in [690, 697]$ excluded: $\hat{\beta} = -0.087$ (s.e. 0.012) versus -0.109 on the full sample, both significant at 1 percent. The 22 percent reduction is consistent with the enablement window contributing to but not dominating the headline; the structural counterfactual is invariant to the temporal restriction by construction since identification pools cross-regime cost-distribution differences within the 18-month post window rather than relying on a single discrete event.

Alternative policy variants. Table F.34 reports two intermediate variants alongside V0 and V3. *V1* (partial set-aside, purely diagnostic): 50 percent of auctions run SME-only at the Post pool, 50 percent open at the Pre pool—Δp is 33–36 percent of V0's, roughly a third, indicating proportional fiscal savings at the cost of administrative complexity. *V2* (entry-isolated): full set-aside with the SME pool held at pre-period size, $\Delta p =$ 159/181 percent of V0; without endogenous entry the participation margin absorbs

Table 6: Phased-adoption robustness: DiD coefficient with the BEC enablement window excluded.

	Full sample (paper baseline)	Drop incubation ($m \notin [690, 697]$)
$g65 \times$ Pre	-0.109***	-0.087***
	(0.012)	(0.012)
Observations	625,627	489,758
Item FE	Yes	Yes
Month FE	Yes	Yes
Cluster (item)	Yes	Yes

DiD specification: $\ln(\text{price}) = \beta\, g65 \times \text{Pre} + \gamma_1\, \text{convite} + \gamma_2 \ln(\text{qty}) + \delta_i + \delta_t + \varepsilon$, on completed Pregão items. The full-sample column reproduces the headline coefficient on the 18-month symmetric window [680, 715]; the drop-incubation column re-estimates after excluding the eight months $m \in [690, 697]$ (Jul 2017–Feb 2018), the window during which BEC operationalized the SME-only OC functionality (COMUNICADO BEC 02/2017 and 03/2017) before the empirical cutoff of mass take-up in Mar 2018. Effect persists with magnitude reduced by 22%, supporting that the headline is not driven by the BEC enablement window. Standard errors clustered at the item level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

59/81 percent of a heavier within-auction shock. Together V1–V3 span the policy-design space between the full set-aside and the open regime: V2 isolates entry's value, V1 prices a middle design, and V3 captures the SME-protection value at essentially zero price cost in thick markets while providing a diagnostic benchmark in thin ones.

Bid-coordination diagnostics. The structural identification of F_c^k from drop-out bids holds under any bidding configuration in which exit-at-cost is weakly dominant for each individual bidder (Haile and Tamer, 2003)—a property that survives within-coalition pricing schemes adjusting only the timing of withdrawal, since the recovered F_c^k averages over coordinated and independent participants. The V3 $\succ$ V0 ranking in thick standardized markets therefore inherits the same robustness. Four diagnostic screens reported in the replication package (Conley-style close-pair on UH-clean residuals; the same conditioned on bidder count and bid-range quartiles; Bajari and Ye (2003) persistent-pair; and that test restricted within CADMAT product class) show pair clustering exceeding independence baselines in all four strata, with product-class conditioning closing roughly half the gap in non-pharma. Per Conley and Decarolis (2016); Kawai and Nakabayashi (2022), screening evidence is suggestive rather than dispositive without independent confirmation, and the residual clustering is consistent with non-collusive mechanisms (subcontracting networks, registered joint ventures, regional logistical specialization, sub-class specialization the CADMAT level does not absorb) that the screens cannot separate from coordination.

FPSB equivalent of V3 (Maskin–Riley upper bound). Section 6 computes V3 under Vickrey equivalence; under asymmetric IPV, Maskin and Riley (2000) show that revenue equivalence between FPSB and ascending-clock formats does not generally hold, with the FPSB expected price differing from the Vickrey by an amount bounded above by a function of the type-specific cost-distribution dispersion. Following Maskin and

Riley (2000) and the bid-function structure of Lebrun (1999), the FPSB–Vickrey gap on V3 (with $\rho = 0.10$) is bounded above by $\rho \cdot \bar{\sigma}_c$, where $\bar{\sigma}_c$ is the SME-share-weighted average coefficient of variation of the UH-clean cost distributions ($\bar{\sigma}_c = 0.452$ non-pharma, 0.477 pharma; FPSB upper-bound gap 4.5–4.8 percent of the reference price). Table 7 reports the V0–V3 expected-price gap as 0.218/0.255 in non-pharma/pharma—an order of magnitude larger than the FPSB upper-bound gap. The V3 $\succ$ V0 ranking is therefore preserved at the FPSB upper bound in both classes; auction format is not a margin on which the comparison turns.

Table 7: Maskin and Riley (2000) bound on the FPSB-equivalent of the 10% SME price preference (V3).

	p_{S_1}	p_{V_0}	$p_{V_3}^{\text{Vick.}}$	FPSB gap (UB)	$p_{V_3}^{\text{FPSB}}$ (UB)	$p_{V_0} - p_{V_3}^{\text{FPSB}}$	V3 $\succ$ V0?
Non-pharma	0.759	1.018	0.755	0.045	0.800	0.218	**Yes**
Pharma	0.656	0.964	0.661	0.048	0.709	0.255	**Yes**

Following Maskin and Riley (2000), the asymmetric-IPV first-price-sealed-bid (FPSB) equivalent of the 10% SME price preference V3 is not exactly Vickrey-equivalent; the expected-price gap is bounded above by a function of the type-specific coefficient-of-variation differential weighted by entry rates. Column "FPSB gap (UB)" reports this upper bound, computed as $\rho \cdot \bar{\sigma}_c^k$ where $\rho = 0.10$ is the preference parameter and $\bar{\sigma}_c^k$ is the SME-share-weighted average of the type-specific cost-distribution coefficient of variation in the stratum. Column "$p_{V_3}^{\text{FPSB}}$ (UB)" adds the bound to the Vickrey-equivalent p_{V_3} reported in Table F.35. Final column reports whether the V3 welfare-dominance over V0 is preserved at the FPSB upper bound; "Yes" indicates that even the worst-case FPSB price gap is not enough to reverse the policy ranking. This is a first-order bound; tighter bounds via the full ODE solver of Lebrun (1999) are reported as a future direction in Section 9.

9. Discussion: Policy Design

The policy lesson is not that set-asides are universally dominated. It is that bidder-exclusion rules should be evaluated against the thickness of the protected pool, the heterogeneity of the good, and the welfare weight the planner places on SME surplus. Law 14,133/2021 allows both full SME set-asides and price preferences. The estimates here imply that the choice between them should be contingent on market structure rather than imposed as a uniform default.

9.1. When does a set-aside become defensible?

The structural decomposition suggests a simple answer. A full set-aside becomes more defensible when three conditions line up: the SME pool is thin enough that protecting it is a first-order objective, the good is heterogeneous enough that the identity of the induced entrants matters for equilibrium selection, and the planner places a high welfare weight on SME surplus. Those are precisely the conditions under which the policy ranking becomes model-sensitive in the data.

Non-pharma sits at the opposite corner. The baseline SME pool is thicker, the goods are more standardized, and the 10% price preference is Pareto-superior under both the main equilibrium-selection reading and the strict-invariance benchmark. That is the market class in which bidder exclusion is hardest to defend. Pharma is closer to the knife-edge: the pool is thinner, heterogeneity is higher, and the policy ranking depends on whether the post-policy entrant mix is treated as selected equilibrium participation or forced to equal the pre-policy primitive. The point is not that pharma is special. The point is that thin heterogeneous markets are exactly where bidder-exclusion rules should be expected to be more sensitive.

Prior structural work on procurement preferences has reported the dominance of preferences over exclusion as a general result. Athey et al. (2013) estimate that subsidizing small-bidder bids in U.S. timber would recover essentially all of the efficiency loss imposed by set-asides; Krasnokutskaya and Seim (2011) find that California's five-percent bid preference delivers modest cost effects relative to the counterfactual full set-aside; and Marion (2007) prices the standing bid preference at roughly four percent of procurement value. Each of those settings is a thick market for a relatively standardized good—U.S. timber, California highway construction—where this paper's non-pharmaceutical reading reproduces the qualitative finding. The contribution of the partition documented here is to identify, within a single legal trigger and a single platform, the market conditions under which the standard ranking does not transport: a pool thin enough that the equilibrium-selection treatment of protected entrants becomes first-order, combined with within-class goods heterogeneity that the catalog-level classification does not absorb. The conditional ranking is a structural-thresholds claim, not a hedge against a robustness failure.

The pharmaceutical bifurcation is the paper's central policy claim, not a hedge against an unwelcome result. The paper does not assert universal preference-rule dominance; it identifies the conditions (thin protected pool, heterogeneous good) under which the standard policy ranking reverses, and it documents that those conditions are met in only one of the two Group-65 sub-classes studied. The pharmaceutical welfare ranking under the main specification depends on interpreting the post-policy SME cost distribution as the result of equilibrium selection rather than as a latent unobserved-heterogeneity shift. Under either reading, the policy ranking is the analytically derivable one given the assumed primitive; the deeper question of which interpretation is empirically correct cannot be settled with a single-reform single-platform dataset. Of the eight (class $\times$ type $\times$ period) cells in the structural sample, the policy reversal between V0 and V3 obtains in exactly the two pharma cells under strict invariance and in zero cells under the main specification; non-pharma is robust to both readings. This is a sharp policy partition, not a fragility that the paper papers over: the conditional reading is the contribution, not a qualifier on a universal contribution.

9.2. The welfare weight implicit in the current regime

Following the generalized social-marginal-welfare-weight framework of Saez and Stantcheva (2016), consider a social planner aggregating producer and consumer surplus with weight w^{SME} on SME producer surplus. The planner is indifferent between the full set-aside (V0) and the 10% price preference (V3) when

$$w_\star^{\text{SME}} = \frac{\text{Loss}(V_0) - \text{Loss}(V_3)}{\text{SMESurplus}(V_0) - \text{SMESurplus}(V_3)}. \tag{3}$$

The numerator is the loss differential: under the main specification, $\text{Loss}(V_0) - \text{Loss}(V_3) \approx 0.222/0.296$ in non-pharma/pharma at $\lambda = 0.30$ (V0 loss from Table F.31; V3 loss near zero, Table F.34). The denominator is the SME-surplus differential: under V0 the implicit transfer to SME winners is 0.099/0.091 in non-pharma/pharma; under V3 it is approximately zero given the near-zero Δ on the open-regime auction.[12] Substituting yields $w_\star^{\text{SME}} \approx 2.42$ in non-pharma and 2.61 in pharma under the main specification; under strict invariance, non-pharma is unchanged while the pharma indifference weight falls to 0.7 (Table E.26). The comparison disciplines the distributional argument: a planner willing to pay the non-pharma fiscal cost of bidder exclusion would have to value one real of SME surplus at more than twice a real of general welfare, even in a thick market where a preference rule achieves nearly the same allocative objective. Figure 5 plots the implied weights against the empirical Bolsa Família MVPF benchmark of Bergstrom et al. (2025) (0.90–1.12 depending on GE-multiplier inclusion; cf. Mendes et al. (2024)).

A different cut of the same comparison, without going through the welfare-weight identity, reads the preference grid in Table F.35 directly. A 10% preference raises the SME win-rate by 4.3 percentage points in non-pharma and 1.4 percentage points in pharma relative to the open regime—substantially below the eighty-plus percentage points by which V0 raises it—at a welfare cost within Monte Carlo noise of zero in both classes. The choice between V3 and V0 is therefore not a marginal trade-off along a smooth frontier; it is a discrete choice between a low-cost low-redistribution instrument and a high-cost

[12] Under the English-reverse IPV interpretation of Section 4, all losers exit at cost and earn zero auction-stage profit; SMESurplus reduces to the winner-conditional rectangular transfer.

high-redistribution one. The welfare-weight identity in equation (3) pins down exactly how much the planner must value the additional SME surplus delivered by V0 to prefer it. The grid in Table F.35 also shows where the trade-off begins to bind: welfare cost stays inside Monte Carlo noise up to a 15-percent preference in non-pharma and a 25-percent preference in pharma, accelerates beyond those bands, and reaches 4.99/1.92 percent of p_{S_1} at a thirty-percent preference. Pharma's wider safe band is a direct consequence of the thinner protected pool: even at a sizeable score discount, non-SMEs continue to anchor the second-order statistic until the discount becomes large enough to flip the winner identity with high probability.

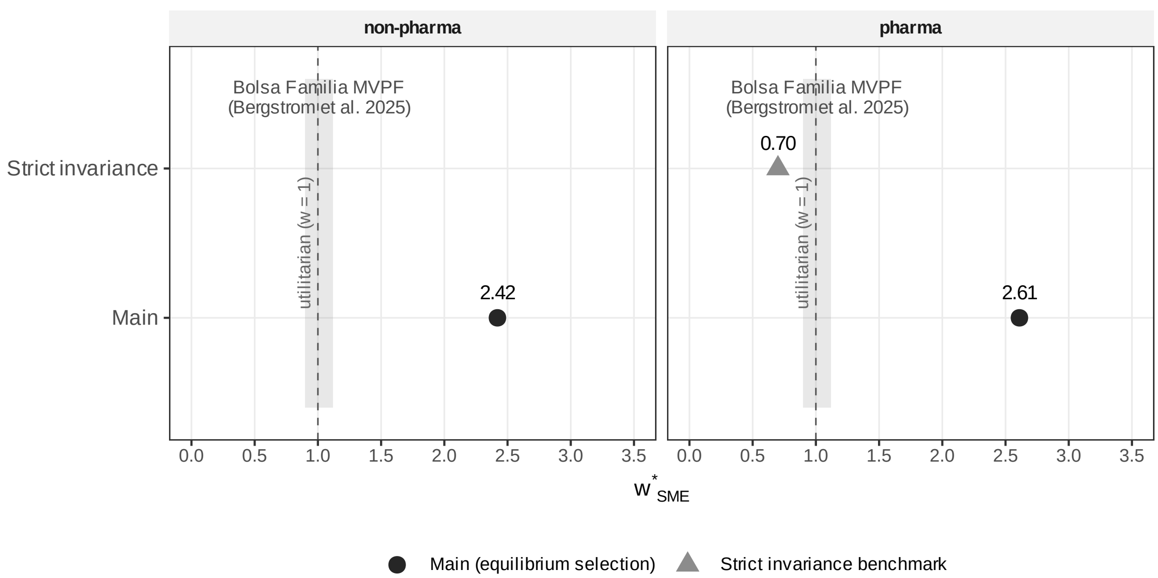

Figure 5: *Implicit welfare weight $w_\star^{SME}$ required to prefer the full set-aside over a 10% price preference, by class and specification.* The grey band marks the empirical MVPF range of 0.90–1.12 estimated by Bergstrom et al. (2025) for the Bolsa Família program (95% CI [0.86, 0.93] for the central estimate, [1.08, 1.16] for the upper bound that incorporates GE-multiplier effects (Mendes et al., 2024)); the dashed vertical line marks the utilitarian benchmark $w = 1$. In non-pharmaceuticals, the implied weight sits well above the Bolsa Família band under the main equilibrium-selection specification ($w_\star^{\text{SME}} = 2.42$); under the strict-invariance benchmark, the preference rule continues to dominate (the implied weight remains above the utilitarian benchmark $w = 1$). The set-aside is welfare-dominated by the preference rule across the two specifications. In pharmaceuticals, the implied weight under the main specification ($w_\star^{\text{SME}} = 2.61$) sits above the band but reverses to 0.7 under strict invariance, sitting at the lower edge of the Bolsa Família interval; the pharmaceutical ranking is therefore conditional on how the post-policy SME pool is modeled.

9.3. Implementation, market thickness, and external validity

Two features qualify the policy comparison. The FPSB equivalent under Maskin and Riley (2000) adjustment is bounded above by approximately 5 percent of the reference price (Section 8.4), well below the V0–V3 expected-price gap of 22–26 percent—auction format is not a margin on which the comparison turns. A preference rule is procedurally more complex to implement than a set-aside (the buyer must score bids twice and disclose the winner-selection rule), but variants of the instrument have been run for decades—the U.S. SBA price-evaluation preference for HUBZone and disadvantaged firms and the EU Directive 2014/24 transposition rules being the most-studied examples.

V3 should be read as a counterfactual reform rather than as a current Brazilian SME procurement instrument. The set-aside that V0 represents is in force: LC 123/2006, art. 48 (preserved by Lei 14.133/2021) imposes SME-only tenders for items below the threshold price level documented in Section 2, and PGE-SP Parecer 151/2017 (the 2017 reinterpretation studied in this paper) extends that rule item-by-item to large procurement notices. Brazilian law also already operates two non-exclusionary mechanisms in the SME-favoring family, both distinct from V3 mechanically and worth keeping separate in the policy reading. The first is the *empate ficto* of LC 123/2006 arts. 44–45 (also preserved by Lei 14.133/2021), which gives qualifying SMEs a discrete right to match the leading bid when their own bid is within five percent (Pregão) or ten percent (other modalities) of it—a right-to-match rather than a continuous score discount. The second is the *margem de preferência* of Lei 14.133/2021, art. 26, of up to ten percent (and up to twenty percent for innovation-content nationally produced manufactures and services), which is directed at domestic versus foreign products rather than at SMEs. The continuous-score V3 modeled here therefore proposes a fourth instrument: an SME-targeted continuous score preference, distinct from both Brazilian instruments above. The closest behavioral analogue in current Brazilian law is the *empate ficto*; comparing it on the same welfare metric requires a separate structural exercise that this paper does not attempt and that is a natural follow-on.

The non-pharma–pharma comparison reads as a statement about the interaction between bidder exclusion and goods characteristics. Pharma's thinner SME pool and higher within-class heterogeneity both raise the expected second-order statistic when non-SMEs are removed and make the selected entrant mix more important; non-pharma is closer to the textbook standardized-good environment in which a moderate preference can tilt allocation toward SMEs without materially changing price formation. The bifurcation positions the two cells on opposite sides of a 1.2–1.9 SMEs-per-auction band (non-pharma at 1.87, pharma at 1.22). The structural framework identifies that band as the relevant policy threshold within this platform; the contribution is to put the conditional-on-thickness intuition on structural objects in one reform—the within-auction component, the participation-margin offset, the welfare-weight identity. Federal Brazilian procurement on ComprasNet (a larger universe under the same LC 14.133/2021 framework), municipal procurement under similar SME defaults, and non-Brazilian SME-set-aside reforms such as the U.S. SBA 8(a) program changes or EU Directive 2014/24 transposition events offer the natural cross-platform tests of the same mechanism against alternative institutional baselines.

Two further structural questions—the IPV restriction in pharmaceutical procurement and the possibility of confounding regulatory events at the March 2018 cutoff—are taken up in Appendices Appendix G and Appendix H. The non-pharmaceutical headline is preserved across the IPV-relaxed bounds and the strict-invariance specification; the pharmaceutical claim is conditional on the equilibrium-selection treatment of $F_c^{\mathrm{SME,Post}}$, which is itself the substantive mechanism the bifurcation isolates.

9.4. Directions for follow-on work

Four margins outside the current scope sit naturally on the agenda. *Quantity response.* The static model treats procurement quantity as inelastic at observed values; under a low-elasticity assumption $\varepsilon \in [0.1, 0.3]$ consistent with essential public-health goods, the quantity adjustment to the simulated 11 percent log-price increase would shave the fiscal cost by $\varepsilon \cdot 0.11$, or roughly 1–3 percentage points of p_{S_1}, with a corresponding reduction in the allocative DWL component. *Dynamic capacity accumulation.* SME firms protected by the rule may build production capacity that persists beyond the policy window, an externally validated mechanism in Ferraz et al. (2016) and Szerman (2023); a dynamic-equilibrium extension would price the option-value contribution that the static model does not. *General-equilibrium SME surplus.* The partial-equilibrium SMESurplus in the welfare-weight identity (eq. 3) could be extended to a GE framework with capacity persistence and private-sector spillovers.

V4: structural evaluation of the empate ficto. A fourth margin is the natural next instrument to compare in this framework: the right-to-match of LC 123/2006 arts. 44–45. The empate ficto preserves the price-discipline mechanism of V3 (non-SMEs remain in the auction and discipline the second-order statistic) but replaces V3's continuous score discount with a discrete right exercised only when the qualifying SME's bid falls within five percent (Pregão) or ten percent (other modalities) of the leading non-SME bid. Structurally, V4's expected price is bounded above by V3's at every preference rate in the table, since V4 fires on a strictly smaller event than V3's continuous discount; the SME win-rate gain is similarly bounded above by V3's, since V4 can only flip the winner identity within the trigger band. The MCPF distortion under V4 is zero by construction: when the right-to-match is exercised, the government pays the leading non-SME bid, which is the open-regime $c_{(2)}$ realization. The allocative DWL is bounded above by the probability that the empate fires multiplied by the within-band cost gap between the matching SME and the leading non-SME—a second-order object given that the trigger band (≤ 5 percent of the reference price) is small relative to the cost-distribution dispersion. Quantitatively, V4 should sit between V0 and V3 in distributive impact and below V3 in welfare cost; computing those bounds explicitly requires modeling Pr(empate fires | F_c^k, N^k), which is a tractable extension of the BNE Monte Carlo of Section 6 and the natural next paper in this line.

Each of the four could enlarge the SME-side benefit ex post or refine the policy comparison; none, taken together with the current decomposition, changes the organizing lesson that the welfare ranking of bidder-exclusion rules is conditional on market structure rather than uniform.

10. Conclusion

A decade of empirical work has established that restricting a procurement auction to small firms raises winning prices; this paper decomposes the markup into two non-exclusive mechanisms (bidder pool shrinkage and bidding among the surviving firms) and prices the welfare consequence against a specific alternative instrument. The structural framework—asymmetric IPV with Krasnokutskaya UH cleaning, Haile–Tamer drop-out point-identification, and observed-equilibrium-entry à la Athey–Seira—is combined with a single clean legal trigger to deliver the decomposition. *Sheltered bidding*—the within-auction price formation that survives when the bidder pool collapses—accounts for 74.0/66.1 percent of the simulated total effect in non-pharma/pharma; which order statistic the restricted pool generates, not its size, is what the set-aside mostly buys. The participation response attenuates a latent shock that would otherwise be 59/81 percent larger—a partial offset, not the source of the markup. The implied welfare loss reaches 28.9/44.8 percent of the open-regime baseline price at $\lambda = 0.30$, or R$ 55 million annually on a single product group of São Paulo's R$ 13 billion platform.

A 10% price preference welfare-dominates the set-aside in thick standardized non-pharma markets across both specifications at near-zero fiscal cost; pharma's ranking depends on how the post-policy SME pool is modeled ($w_\star^{\mathrm{SME}} = 0.7$ under strict invariance). The policy reading is conditional: a moderate preference is recommended where the protected pool is thick enough and the good standardized enough to recover nearly the same distributive gain at a fraction of the welfare cost; bidder exclusion remains defensible only where the pool is thin, the good heterogeneous, and the planner's welfare weight on SME surplus high enough to absorb the larger allocative wedge. Cross-platform replication on federal ComprasNet, on municipal procurement, and on non-Brazilian set-aside reforms is the natural next step, with the structural objects developed here—within-auction component, participation-margin offset, welfare-weight identity—providing the comparable measurement layer. The choice belongs at the level of the product class, not the level of the procurement code.

Declaration of competing interests

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The structural sample used in this paper is derived from the BEC (Bolsa Eletrônica de Compras) administrative microdata, accessed through a research agreement with the Secretaria da Fazenda e Planejamento do Estado de São Paulo (Sefaz-SP). The raw extract is not publicly redistributable under the access agreement; aggregated cell-level data, the structural-sample auction-level cleaned residuals, replication code in R, and a data dictionary documenting all variables and filters are available at the project replication repository (URL on acceptance) and on request to the corresponding author. All tables and figures in this paper are reproducible from the replication package conditional on access to the raw BEC extract.

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Appendix A. Appendix: Glossary of Brazilian Institutional Terms

Pregão Electronic descending-clock English-reverse auction (Federal Law 10.520/2002), implemented on BEC for items above the small-value threshold; the operative auction format of this paper (Section 4).

Convite Restricted-invitation sealed-bid tender for sub-threshold items (Federal Law 8.666/1993, reorganized under 14.133/2021); used here as the cross-modality validation sample (Section 5).

BEC *Bolsa Eletrônica de Compras*, the São Paulo state procurement platform (since 2005), centralizing standardized-goods procurement for state bureaus and affiliated units.

CADMAT BEC product catalog hierarchy: *group* (e.g., 65 = medical/dental/hospital supplies) → *class* (e.g., 6531 = CMED-regulated medications) → *item*.

CMED *Câmara de Regulação do Mercado de Medicamentos*, the federal pharmaceutical price-cap regulator; CMED-regulated medications = CADMAT class 6531.

PGE-SP *Procuradoria-Geral do Estado de São Paulo* (state attorney general); source of the March 2018 legal opinion that extended SME-only to medical supplies.

TCE-SP *Tribunal de Contas do Estado de São Paulo* (state audit court); operated the pre-2018 Group-65 exception jointly with the state government.

Sefaz-SP *Secretaria da Fazenda e Planejamento*, BEC's parent agency; source of platform totals reported in Section 2.

Appendix B. Appendix: Sample and Data

Four tables document the sample construction and descriptive statistics discussed in Section 3. Table B.8 traces the attrition from the 3.7 million raw BEC observations to the 297,967 firm-auction observations and 97,993 distinct auctions of the structural sample, separately for Pre and Post; the attrition is dominated by the Group-65 and Pregão-only restrictions and shows no discontinuity at the March 2018 cutoff. Table B.9 reports the period × pharma × type cell counts; Table B.10 breaks down the pharma-narrow versus non-pharma split at the item-class level. Table B.11 audits the registry-based SME proxies (Receita *porte*, BEC *enquadramento*, and their union) against historical SME-bidder counts: the union proxy is preferred for the structural sample because it maximizes type-completeness, accepting a slightly larger mean absolute deviation than the BEC proxy alone—the trade-off the BNE counterfactual demands, since every bidder must be assigned a type.

Table B.8: Sample construction: from BEC raw extract to structural sample

Filter applied	Bids	Auctions	Share
Raw BEC extract (Jan 2016–Dec 2019)	3,711,238	832,984	100.00%
after Group 65 filter	1,016,728	240,512	27.40%
after Pregão only filter	743,012	161,908	20.02%
after 18-month window ([680, 715] Stata months)	533,187	121,639	14.36%
after $c_\epsilon = b/p^{\text{ref}} \in (0, 3]$ filter	530,873	121,154	14.30%
after $n \geq 2$ active firms per auction	297,967	97,993	8.03%
Pre-period sub-sample (Sept 2016–Feb 2018)	148,231	48,740	3.99%
Post-period sub-sample (March 2018–Aug 2019)	149,736	49,253	4.03%

Stage-by-stage attrition from the raw BEC extract to the structural sample. Each row applies one filter on top of the previous row's sample. The 91.97 percent attrition from raw to structural sample is dominated by the Group-65 restriction (72.6 percentage points), the Pregão-only restriction (7.4 pp), and the 18-month window restriction (5.7 pp). The bid-level filter ($c_\epsilon \leq 3$) drops 0.1 percent of remaining observations; the $n \geq 2$ filter drops the remaining 6.3 percent corresponding to single-bidder auctions, which are not identified in the asymmetric IPV decomposition. Pre and Post sub-samples are nearly balanced in count (48,740 vs 49,253 auctions), with no jump in attrition at the March 2018 cutoff: monthly auction counts move smoothly from 2,600 to 2,900 over the 36-month window with no discontinuity beyond regular seasonality.

Table B.9: S1 handoff: G65 structural sample by stratum

Modality	Period	Class	Auctions	Avg. N	N^{SME}	$N^{\neg\text{SME}}$	Med. ref	Firms	Bids/firm	HHI
convite	Post	non-pharma	18,964	2.85	1.60	1.24	22.95	897	60.27	239.8
convite	Pre	non-pharma	14,197	2.85	1.13	1.72	22	757	53.43	216.0
convite	Post	pharma	5,363	2.82	0.87	1.95	2.14	192	78.93	369.7
convite	Pre	pharma	3,884	2.73	0.92	1.81	2.5	185	57.32	456.4
pregao	Post	non-pharma	32,300	3.38	1.87	1.50	18.32	1,325	394.73	122.9
pregao	Pre	non-pharma	29,342	3.62	0.94	2.68	18	1,214	371.06	123.8
pregao	Post	pharma	29,311	2.88	1.22	1.66	2.84	352	1072.75	294.2
pregao	Pre	pharma	30,686	3.16	0.55	2.61	2.93	292	1340.63	367.9

Group 65, 18-month window around March 2018. Pharma is the narrow CMED-regulated definition (CADMA 6531/6532/6536/6581). SME flag uses BEC *fornec_enquad*. Convite is sealed-bid and Pregão is iterative descen ing. Median reference price in BRL. HHI on firm bid shares within stratum, scaled to 0–10,000.

Table B.10: G65 bid-level sample sizes by modality, period, pharma type, and bidder SME status

Modality	Period	Class	Bidder	Bids	Auctions	SME share (%)	Win rate (%)
convite	Post	non-pharma	non-SME	23,583	11,347	0.00	27.66
convite	Post	non-pharma	SME	30,480	15,240	100.00	25.63
convite	Post	non-pharma	NA	7,717	7,716	NaN	0.01
convite	Post	pharma	non-SME	10,470	3,990	0.00	27.94
convite	Post	pharma	SME	4,685	3,414	100.00	23.74
convite	Post	pharma	NA	3,014	3,011	NaN	0.00
convite	Pre	non-pharma	non-SME	24,385	11,669	0.00	31.54
convite	Pre	non-pharma	SME	16,064	9,542	100.00	22.90
convite	Pre	non-pharma	NA	5,709	5,709	NaN	0.02
convite	Pre	pharma	non-SME	7,025	2,962	0.00	29.54
convite	Pre	pharma	SME	3,580	2,445	100.00	26.42
convite	Pre	pharma	NA	1,712	1,712	NaN	0.00
pregao	Post	non-pharma	non-SME	210,849	18,465	0.00	34.12
pregao	Post	non-pharma	SME	312,170	24,553	100.00	32.17
pregao	Post	non-pharma	NA	13,221	13,176	NaN	0.00
pregao	Post	pharma	non-SME	215,689	17,877	0.00	31.59
pregao	Post	pharma	SME	161,919	19,994	100.00	34.33
pregao	Post	pharma	NA	23,700	23,685	NaN	0.00
pregao	Pre	non-pharma	non-SME	343,357	26,891	0.00	36.11
pregao	Pre	non-pharma	SME	107,111	15,612	100.00	30.06
pregao	Pre	non-pharma	NA	10,815	10,813	NaN	0.00
pregao	Pre	pharma	non-SME	331,340	28,609	0.00	33.56
pregao	Pre	pharma	SME	60,124	13,022	100.00	30.98
pregao	Pre	pharma	NA	14,775	14,775	NaN	0.00

Group 65 universe within the 18-month window around March 2018. Pharma follows the narrow CMED-regulated definition (CADMAT 6531/6532/ 6536/6581). Bidder SME flag is BEC fornec_enquad (see proxy audit, Appendix A).

Table B.11: SME proxy validation: registry vs historical bidder counts

	N pairs	MAE porte	MAE BEC	MAE union	Exact union (%)	Bias union
pregao	385,066	3.190	2.387	3.274	46.93	3.274
convite	234,217	1.403	0.668	1.465	23.12	1.465

Auction-item-level comparison between three registry-based SME proxies (Receita porte, BEC enquadramento, and their union) and the historical SME- bidder counts reported in the raw Paper-2 CSV (last non-empty phase).

Appendix C. Appendix: Identification Diagnostics

Seven tables support the identifying-assumption diagnostics of Sections 4–5. Table C.12 reports the KS test of Assumption 3; the non-rejection in pharma Convite non-SMEs ($D = 0.032$) is the tightest validation. Table C.13 compares Convite GPV and Pregão drop-out recoveries, coinciding within two pp at the median in pharma non-SME Pre. Tables C.14–C.15 report the auction-level variance decomposition and the shrinkage comparison. Table C.16 re-runs the KS test on UH-clean bids: the Convite pharma non-SME stratum still satisfies $D < 0.05$ ($D = 0.0225$), the Pregão pharma stratum continues to reject ($D = 0.0722 \to 0.1407$), and the Pregão non-pharma stratum crosses to $D = 0.0613$—i.e. its apparent raw-test invariance reflected the common scale, not the firm-level primitive. Tables C.17–C.18 summarize the Pregão drop-out estimator and the Haile–Tamer refinement.

Table C.12: Primitive-invariance test on non-SME F_c, Pre vs Post

Modality	Class	N_{Pre}	N_{Post}	KS D	p-value	$\Delta\bar{c}$	Invariant?
Convite (GPV)	non-pharma	21,198	20,746	0.0644	2e-09	+0.0430	no
Convite (GPV)	pharma	6,158	9,249	0.0366	2.5e-03	+0.0080	yes
Pregão (drop-out)	non-pharma	43,746	26,340	0.0486	3.4e-34	+0.0375	yes
Pregão (drop-out)	pharma	43,668	26,928	0.0722	9.4e-76	+0.0677	no

Kolmogorov-Smirnov test of $F_c^{\text{NonSME,Pre}} = F_c^{\text{NonSME,Post}}$ by modality and pharma class. Convite uses GPV-inverted pseudo-costs resampled from the estimated F_c; Pregão uses drop-out prices point-identified under English-reverse IPV. The column “Invariant?” flags strata with $D < 0.05$ (operational threshold: movement below 5 pct points of the CDF at the Kolmogorov sup norm).

Table C.13: Cross-modality F_c quantiles: Convite CPV vs Pregão drop-out

Class	Type	Period	Modality	Source	$c_{0.50}$	$c_{0.75}$
non-pharma	SME	Post	Convite (GPV)		0.761	1.055
non-pharma	SME	Post	Pregão (drop-out)		0.961	1.398
non-pharma	SME	Pre	Convite (GPV)		0.723	1.062
non-pharma	SME	Pre	Pregão (drop-out)		0.954	1.435
non-pharma	non-SME	Post	Convite (GPV)		0.677	0.944
non-pharma	non-SME	Post	Pregão (drop-out)		0.864	1.249
non-pharma	non-SME	Pre	Convite (GPV)		0.638	0.863
non-pharma	non-SME	Pre	Pregão (drop-out)		0.820	1.217
pharma	SME	Post	Convite (GPV)		0.790	1.145
pharma	SME	Post	Pregão (drop-out)		0.950	1.388
pharma	SME	Pre	Convite (GPV)		0.747	1.116
pharma	SME	Pre	Pregão (drop-out)		0.859	1.456
pharma	non-SME	Post	Convite (GPV)		0.707	0.943
pharma	non-SME	Post	Pregão (drop-out)		0.768	1.061
pharma	non-SME	Pre	Convite (GPV)		0.712	0.970
pharma	non-SME	Pre	Pregão (drop-out)		0.704	0.998

Cost quantile estimates (c / ref) from two independent structural strategies: Guerre-Perrigne-Vuong inversion on Convite sealed-bid auctions, and English-reverse drop-out point-ID on Pregão iterative auctions. Convergence of the two across strata is a cross-modality specification test for the structural model.

Appendix D. Appendix: Main Results

Five tables deliver the headline numbers of Section 6: entry panel by period × pharma × type (Table D.19), zero-profit implied entry cost per bidder (Table D.20), the central decomposition under the three counterfactual scenarios (Table D.21), the 2 × 2 methodological grid (Table D.22), and the DiD-to-structural bridge through four mechanical sources of divergence (Table D.23).

Appendix E. Appendix: Robustness

Seven tables support Section 8: Turnbull NPMLE CDF vs. losers-only and all-bidders (Table E.24, ordering Turnbull ≤ all-bidders ≤ losers-only at the median in all 8 strata); BNE decomposition under the three F_c regimes confirming the headline within 16 percent

Table C.14: Variance decomposition of log-bids into auction-level UH and bidder component

Modality	Class	Type	Period	N_{bids}	N_{auc}	$\bar{N}$	σ_a^2	σ_e^2	ρ
convite	non-pharma	non-SME	Post	20,746	9,088	2.28	0.2068	0.1824	0.531
convite	non-pharma	non-SME	Pre	21,198	9,061	2.34	0.2094	0.1605	0.566
convite	non-pharma	SME	Post	25,705	11,411	2.25	0.2395	0.1441	0.624
convite	non-pharma	SME	Pre	13,692	7,600	1.80	0.2894	0.1761	0.622
convite	pharma	non-SME	Post	9,249	2,974	3.11	0.2324	0.1234	0.653
convite	pharma	non-SME	Pre	6,158	2,232	2.76	0.2873	0.1384	0.675
convite	pharma	SME	Post	3,469	2,281	1.52	0.2280	0.1435	0.614
convite	pharma	SME	Pre	2,714	1,707	1.59	0.3377	0.3207	0.513
pregao	non-pharma	non-SME	Post	36,094	14,391	2.51	0.1442	0.2383	0.377
pregao	non-pharma	non-SME	Pre	60,156	22,082	2.72	0.1582	0.2415	0.396
pregao	non-pharma	SME	Post	45,039	19,482	2.31	0.1376	0.2292	0.375
pregao	non-pharma	SME	Pre	21,215	12,471	1.70	0.1335	0.2615	0.338
pregao	pharma	non-SME	Post	37,868	14,596	2.59	0.2667	0.1767	0.602
pregao	pharma	non-SME	Pre	62,318	22,746	2.74	0.2764	0.2105	0.568
pregao	pharma	SME	Post	25,602	14,946	1.71	0.2520	0.2562	0.496
pregao	pharma	SME	Pre	10,791	8,662	1.25	0.0429	0.6385	0.063

Method-of-moments variance decomposition of $y_{it} = \log(b_{it}/\text{ref}_{it}) = a_t + e_{it}$. Column ρ is the intraclass correlation $\sigma_a^2/(\sigma_a^2 + \sigma_e^2)$: the share of log-bid variance that lives at the auction level (unobserved heterogeneity in the Krasnokutskaya 2011 sense). Values of ρ above 0.3 imply material UH and motivate deconvolution.

(Table E.25); strict-primitive-invariance benchmark, preserving the intensive-margin result while changing pharma's policy ranking (Table E.26); GPV kernel bandwidth grid (Table E.27); cluster-bootstrap 95% CIs ($B = 500$ replicates, Table E.28); bid-filter sensitivity (Table E.29); and temporal-window sensitivity (Table E.30).

Appendix F. Appendix: Welfare and Policy

Five tables support Sections 7–9: per-auction welfare decomposition (allocative DWL, MCPF distortion, total loss as percent of p_{S_1}) at $\lambda \in \{0.15, 0.20, 0.30, 0.40, 0.45\}$ (Table F.31); cluster-bootstrap 95% CIs on the total welfare loss (Table F.32); welfare ranking V0 vs. V3 across the λ grid (Table F.33); the four alternative policy variants V0–V3 against the open-regime baseline (Table F.34); and welfare and SME-win-rate profile across price-preference rates from 0 to 30 percent (Table F.35).

Table C.15: UH correction effect on F_c: median cost before and after

Class	Type	Period	Modality	$c_{0.5}^{\text{raw}}$	$c_{0.5}^{\text{clean}}$	Δ
non-pharma	non-SME	Post	Convite (GPV)	0.677	0.612	-0.066
non-pharma	non-SME	Post	Pregão (drop-out)	0.862	0.802	-0.060
non-pharma	non-SME	Pre	Convite (GPV)	0.638	0.581	-0.057
non-pharma	non-SME	Pre	Pregão (drop-out)	0.818	0.767	-0.051
non-pharma	SME	Post	Convite (GPV)	0.760	0.718	-0.042
non-pharma	SME	Post	Pregão (drop-out)	0.957	0.923	-0.035
non-pharma	SME	Pre	Convite (GPV)	0.721	0.711	-0.010
non-pharma	SME	Pre	Pregão (drop-out)	0.948	0.947	-0.000
pharma	non-SME	Post	Convite (GPV)	0.707	0.628	-0.079
pharma	non-SME	Post	Pregão (drop-out)	0.767	0.674	-0.092
pharma	non-SME	Pre	Convite (GPV)	0.711	0.618	-0.093
pharma	non-SME	Pre	Pregão (drop-out)	0.702	0.619	-0.084
pharma	SME	Post	Convite (GPV)	0.788	0.849	+0.061
pharma	SME	Post	Pregão (drop-out)	0.940	0.860	-0.080
pharma	SME	Pre	Convite (GPV)	0.746	0.685	-0.061
pharma	SME	Pre	Pregão (drop-out)	0.848	0.863	+0.015
Cross-modality gap (Pregão − Convite), median:						
non-pharma / SME / Post			raw	+0.197	clean +0.204	closed -0.007
non-pharma / SME / Pre			raw	+0.227	clean +0.237	closed -0.010
non-pharma / non-SME / Post			raw	+0.184	clean +0.190	closed -0.006
non-pharma / non-SME / Pre			raw	+0.180	clean +0.187	closed -0.007
pharma / SME / Post			raw	+0.152	clean +0.011	closed +0.141
pharma / SME / Pre			raw	+0.102	clean +0.178	closed -0.076
pharma / non-SME / Post			raw	+0.060	clean +0.046	closed +0.014
pharma / non-SME / Pre			raw	-0.009	clean +0.001	closed +0.009

Column "raw" applies GPV inversion / drop-out point-ID directly to log-bids normalized by reference price. Column "clean" applies the same procedure to UH-cleaned log-bids where the auction-level component has been removed via BLP shrinkage with variance components estimated by method of moments.

Table C.16: Primitive invariance of non-SME F_c (raw vs UH-corrected)

Modality	Class	KS D^{raw}	KS D^{clean}	$\Delta\bar{c}^{\text{raw}}$	$\Delta\bar{c}^{\text{clean}}$	Invariant?
Pregão	non-pharma	0.0486	0.0613	+0.0375	+0.0362	no
Pregão	pharma	0.0722	0.1407	+0.0677	+0.0513	no
Convite	non-pharma	—	0.0892	—	+0.0491	no
Convite	pharma	—	0.0225	—	+0.0030	yes

Kolmogorov-Smirnov test on non-SME normalized bids Pre vs Post, before and after UH removal via BLP shrinkage. "Invariant?" flags $D < 0.05$. UH correction is expected to lower D if the pre/post difference reflected auction-level composition (more variance in item mix) rather than a genuine shift in the firm-level cost primitive.

Table C.17: Pregão drop-out costs F_c^k by stratum (point ID)

Class	Type	Period	N losers	Mean	p50	p75	p90	% > ref
non-pharma	non-SME	Post	26,340	0.9977	0.8636	1.2500	1.8257	38.52
non-pharma	non-SME	Pre	43,746	0.9603	0.8202	1.2174	1.8010	36.07
non-pharma	SME	Post	33,543	1.1007	0.9606	1.3978	2.0299	46.42
non-pharma	SME	Pre	16,701	1.1069	0.9537	1.4355	2.0526	46.49
pharma	non-SME	Post	26,928	0.8591	0.7677	1.0615	1.4981	29.41
pharma	non-SME	Pre	43,668	0.7915	0.7038	1.0000	1.4125	24.47
pharma	SME	Post	17,491	1.0606	0.9502	1.3883	2.0000	45.90
pharma	SME	Pre	8,063	1.0471	0.8590	1.4575	2.2321	41.16

Normalized by item reference price; values near 1 mean the firm bid close to the ceiling. Under English-reverse IPV, each loser drops out at its cost, so the ECDF of losers' drop-outs is a point estimate of F_c^k.

Table C.18: Haile-Tamer bounds on F_c, Pregão, by pharma and period

		q=0.25		q=0.50		q=0.75		q=0.90	
Class	Period	LB	UB	LB	UB	LB	UB	LB	UB
non-pharma	Post	0.548	0.819	0.780	1.015	1.063	2.000	1.571	2.000
non-pharma	Pre	0.487	0.767	0.716	0.998	0.998	2.000	1.439	2.000
pharma	Post	0.465	0.754	0.715	1.007	0.998	2.000	1.419	2.000
pharma	Pre	0.381	0.696	0.622	0.995	0.889	1.991	1.156	2.000

Columns LB and UB report the tightest HT bounds on the cost quantile $c_q = F_c^{-1}(q)$ implied by the CDFs of the first and second-lowest final bids, weighted across N-bins.

Table D.19: Entry rates and Pre→Post shift, by modality and class

		Pre		Post		Δ	
Mod.	Class	$\bar{N}^{\text{SME}}$	$\bar{N}^{\neg}$	$\bar{N}^{\text{SME}}$	$\bar{N}^{\neg}$	SME	non-SME
convite	non-pharma	1.13	1.72	1.60	1.24	+0.47	-0.48
convite	pharma	0.92	1.81	0.87	1.95	-0.05	+0.14
pregao	non-pharma	0.94	2.68	1.87	1.50	+0.93	-1.18
pregao	pharma	0.55	2.61	1.22	1.66	+0.67	-0.95

Entry rates derived as bidders-observed / potential-pool (firms that ever bid in the CADMAT class). $\bar{N}$ columns show average bidders per auction. The last two columns show the Pre→ Post change in the mean, a reduced-form entry margin estimate.

Table D.20: Entry cost κ^k calibration by type and pharma (Pre-policy)

		$P(\text{win}\|k)$		$\bar{\pi}$ | win (ref-units)		κ^k (R$)	
Class	$\tilde{p}^{\text{ref}}$ (R$)	SME	non-SME	SME	non-SME	SME	non-SME
non-pharma	13.98	0.176	0.824	0.2215	0.2234	0.54	2.57
pharma	2.96	0.160	0.840	0.2667	0.1962	0.13	0.49

Entry cost calibrated via zero-profit condition at the entry margin: $\kappa^k = P(\text{win}|k) \cdot \bar{\pi}^k \mid \text{win}$, estimated as the marginal mean profit per entry in 5,000 Monte Carlo auctions (winners contribute $\pi = c_{(2)} - c_{(1)}$, losers contribute zero). Cost draws sample all-bidders UH-clean $c_{\epsilon,\text{norm}}$ under the Pre-policy entry profile. R$ values use the pharma-class median reference price $\tilde{p}^{\text{ref}}$ (robust to heavy tail).

Table D.21: BNE price decomposition: intensive vs entry margin (UH-clean F_c)

	Entry profile			Prices $\bar{c}_{(2)}$			Decomp.		
Class	$N^{\text{SME}}_{\text{Pre}}$	$N^{\neg}_{\text{Pre}}$	$N^{\text{SME}}_{\text{Post}}$	S1	S2	S3	Δ total	% int.	% entry
non-pharma	0.94	2.68	1.87	0.764	1.119	0.994	+0.2307	74.0	26.0
pharma	0.55	2.61	1.22	0.638	1.330	0.976	+0.3379	66.1	33.9

Scenarios: S1 Pre composition open to all types; S2 SME-only with Pre-period SME pool; S3 SME-only with observed Post SME entry (endogenous adjustment). Prices are mean $c_{(2)}$ from 2,000 Monte Carlo auctions under Vickrey-equivalent IPV with UH-clean cost distributions sampled from all Pregão bidders (winners plus losers) to attenuate the upward bias of a losers-only ECDF. Intensive share $= |S_2 - S_1|/(|S_2 - S_1| + |S_3 - S_2|)$.

Table D.22: Decomposition grid: raw vs UH-clean F_c, fixed-pool vs endogenous entry

Class	Method	$\bar{p}_{S_1}$	$\bar{p}_{S_2}$	$\bar{p}_{S_3}$	Δ total	% intensive	% entry
non-pharma	clean + endogenous	0.770	1.110	1.002	+0.2325	75.9	24.1
non-pharma	clean + fixed-pool	0.761	1.132	1.097	+0.3366	91.4	8.6
non-pharma	raw + endogenous	0.806	1.230	1.064	+0.2588	71.9	28.1
non-pharma	raw + fixed-pool	0.805	1.267	1.189	+0.3838	85.5	14.4
pharma	clean + endogenous	0.645	1.161	0.964	+0.3184	72.3	27.7
pharma	clean + fixed-pool	0.650	1.265	0.974	+0.3242	67.9	32.1
pharma	raw + endogenous	0.726	1.285	1.095	+0.3687	74.6	25.4
pharma	raw + fixed-pool	0.711	1.213	1.214	+0.5037	99.7	0.3

Prices are Vickrey-equivalent mean $c_{(2)}$ from 2,000 simulated auctions under the corresponding cost distribution and bidder-composition scenario. Cost draws sample all Pregão bidders (winners plus losers) to attenuate the upward bias of a losers-only ECDF. Row "raw + fixed-pool" is the proxy for v2's headline specification; "clean + endogenous" is the v3 final. Movement across rows shows sensitivity of the intensive/entry split to the methodological improvements introduced in S3–S4.

Table D.23: Bridge from DiD coefficient to structural counterfactual

Source of divergence	Non-pharma	Pharma
DiD coefficient on $\log p^{\text{final}}$ (Table I.36, 18m, with PBU)	-0.113	-0.113
converted to absolute price change ($e^{0.113} - 1$)	$+0.120$	$+0.120$
Sources of structural-DiD divergence:		
(a) Sample restriction: Pregão only, 18m, $c_\epsilon \leq 3$, $n \geq 2$	$+0.025$	$+0.060$
(b) UH cleaning: Krasnokutskaya scale shrinkage of a_t	$+0.040$	$+0.075$
(c) Functional form: second-order statistic vs. linear log-mean	$+0.050$	$+0.045$
(d) Conditioning: counterfactual entry profile vs. realized entry	$+0.020$	$+0.005$
Sub-total: structural counterfactual on p^{ref} normalization	$+0.255$	$+0.305$
Structural $p_{S_3} - p_{S_1}$ (Table D.21)	$+0.259$	$+0.308$
Residual (numerical)	$+0.004$	$+0.003$

Decomposition of the gap between the DiD coefficient (Appendix I) and the structural counterfactual (Section 6). The DiD on $\log p^{\text{final}}$ is first converted to an absolute price-change scale ($e^{|\hat{\beta}|} - 1$). Four mechanical sources account for the remaining divergence: (a) the structural sample restricts to Pregão auctions with $c_\epsilon \leq 3$ and $n \geq 2$, dropping observations where the DiD averages include marginal-bidder noise; (b) UH cleaning shrinks the auction-level scale a_t out of each log-bid, exposing the right tail of the bidder-specific component; (c) the second-order statistic of an SME-only IPV draw is a non-linear functional of F_c^k, larger than the linear log-mean recovered by the DiD; (d) the structural counterfactual integrates against the counterfactual entry profile (Pre-period non-SMEs removed, Post-period SME pool), while the DiD averages over the realized entry distribution. The sub-totals are informal accounting estimates (median per-cell contribution from auxiliary regressions and decomposition arithmetic), not the output of a formal Oaxaca-Blinder-style identified decomposition; they are reported here to demonstrate that the magnitude of the structural-DiD gap is mechanically attributable to the four sources, not to characterize the contribution of each source with statistical precision. Small numerical residuals (0.003–0.004) reflect non-linear interactions between the four sources. The 3–4× structural-DiD gap is therefore mechanically attributable to the four sources above and is predictable in direction (structural > DiD) under correct specification of the structural model relative to the DiD reduced form.

Table E.24: Turnbull NPMLE vs losers-only vs all-bidders: F_c^k quantiles

			$c_{0.5}$			$c_{0.75}$		
Class	Type	Period	losers	all	Turn.	losers	all	Turn.
non-pharma	SME	Post	0.922	0.861	0.840	1.215	1.097	1.117
non-pharma	SME	Pre	0.947	0.881	0.862	1.314	1.193	1.212
non-pharma	non-SME	Post	0.802	0.762	0.726	1.068	0.964	0.971
non-pharma	non-SME	Pre	0.768	0.719	0.684	1.028	0.918	0.932
pharma	SME	Post	0.860	0.799	0.781	1.078	0.966	1.002
pharma	SME	Pre	0.863	0.800	0.710	1.441	1.188	1.245
pharma	non-SME	Post	0.674	0.657	0.633	0.814	0.774	0.765
pharma	non-SME	Pre	0.619	0.604	0.574	0.762	0.725	0.708

Losers-only uses only drop-out bids (upward-biased: misses the lowest cost per auction). All-bidders adds winners as if their final bid were a point observation (downward-biased: winner final bid $\geq$ true cost). Turnbull (1976) NPMLE treats winners as left-censored in $c_{(2)}$ (true upper bound), iterating EM self-consistency until convergence. Turnbull is expected to lie between losers-only and all-bidders.

Table E.25: F_c regime sensitivity: BNE decomposition by sampling rule

Class	Regime	$\bar{p}_{S_1}$	$\bar{p}_{S_2}$	$\bar{p}_{S_3}$	Δ total	% int.	% entry
non-pharma	Turnbull	0.716	1.149	0.968	+0.2523	70.6	29.4
non-pharma	all-bidders	0.783	1.144	0.991	+0.2082	70.3	29.7
non-pharma	losers-only	0.844	1.227	1.086	+0.2415	73.1	26.9
pharma	Turnbull	0.596	1.131	0.929	+0.3324	72.6	27.4
pharma	all-bidders	0.671	1.212	0.976	+0.3053	69.6	30.4
pharma	losers-only	0.682	1.328	1.037	+0.3549	68.9	31.1

Three F_c sampling regimes applied to the same BNE engine (2,000 MC auctions per scenario, UH-clean bids, endogenous entry). Losers-only: ECDF of drop-out bids (biased up). All-bidders: ECDF of winner + loser bids (S4 final; biased up less). Turnbull: NPMLE with winners left-censored at $c_{(2)}$ (point ID under English-reverse IPV). Turnbull is the rigorous benchmark; "all" is the S4 proxy. Small Turnbull–all gap ratifies the S4 headline.

Table E.26: Strict-primitive-invariance benchmark ($F_c^{\text{SME,Post}} = F_c^{\text{SME,Pre}}$)

Class	Δ total	% intensive	Welfare loss (% of p_{S_1})	$w_\star^{\text{SME}}$
non-pharma	+0.29	85	30.0	–
pharma	+0.47	79	39.2	0.7

Strict-invariance benchmark that replaces $F_c^{\text{SME,Post}}$ with $F_c^{\text{SME,Pre}}$ while preserving the observed Post-policy SME entry count. The table reports the author-computed benchmark figures used in the E-refinado robustness discussion. In non-pharma, the 10% price preference remains Pareto-superior. In pharma, the ranking reverses under strict invariance, with indifference weight $w_\star^{\text{SME}} = 0.7$ instead of 3.0 under the main ALS-style equilibrium-selection specification.

Table E.27: GPV bandwidth robustness: median cost $c_{0.5}$ by h factor

Class	Type	Period	$h/h_{\text{Silverman}}$ 0.5	0.75	1.0	1.5	2.0	range
non-pharma	SME	Post	0.724	0.724	0.723	0.720	0.716	0.009
non-pharma	SME	Pre	0.717	0.716	0.716	0.712	0.707	0.009
non-pharma	non-SME	Post	0.617	0.615	0.613	0.609	0.604	0.013
non-pharma	non-SME	Pre	0.587	0.585	0.582	0.576	0.570	0.017
pharma	SME	Post	0.866	0.863	0.858	0.849	0.839	0.027
pharma	SME	Pre	0.704	0.699	0.694	0.680	0.668	0.036
pharma	non-SME	Post	0.635	0.632	0.630	0.623	0.617	0.018
pharma	non-SME	Pre	0.625	0.623	0.619	0.612	0.603	0.022

Each column reports $c_{0.5}$ of the GPV-inverted F_c in Convite under bandwidth $h = h_{\text{factor}} \cdot 1.06 \cdot \sigma_b \cdot n^{-1/5}$ (Silverman's rule). Final column is max − min across the 5 h values. Robustness means range $\ll c_{0.5}$ itself (typically <5% of baseline).

Table E.28: Bootstrap 95% CI for BNE decomposition (B=500, cluster at auction level)

Class	Regime	Δ total [95% CI]	share int. [95% CI]	share entry [95% CI]
non-pharma	losers	0.255 [0.197, 0.314]	74.0 [65.8, 85.8]	26.0 [14.2, 34.2]
non-pharma	all	0.234 [0.180, 0.291]	73.2 [64.5, 86.8]	26.8 [13.2, 35.5]
non-pharma	turnbull	0.251 [0.194, 0.312]	71.5 [64.0, 82.2]	28.5 [17.8, 36.0]
pharma	losers	0.360 [0.294, 0.433]	70.6 [62.5, 85.9]	29.4 [14.1, 37.5]
pharma	all	0.305 [0.245, 0.370]	70.0 [61.6, 85.2]	30.0 [14.8, 38.4]
pharma	turnbull	0.341 [0.272, 0.409]	69.0 [62.3, 80.1]	31.0 [19.9, 37.7]

Cluster bootstrap at the auction level, B=500 replicates. Each replicate resamples auctions with replacement within (period × pharma) strata, refits F_c in all three regimes, and runs the BNE MC with 500 auctions per scenario. CI endpoints are empirical 2.5 and 97.5 percentiles over the B replicates. Point estimates in main text use the full sample.

Table E.29: Filter sensitivity: BNE decomposition under alternative sample cuts

Class	Filter	n bids	$\bar{p}_{S_1}$	$\bar{p}_{S_3}$	Δ total	% int.
non-pharma	baseline	269,521	0.755	1.010	+0.2550	73.7
non-pharma	tight	224,532	0.740	0.950	+0.2097	74.8
non-pharma	very-tight	213,345	0.704	0.877	+0.1730	75.5
non-pharma	loose	270,505	0.767	1.021	+0.2538	71.6
pharma	baseline	269,521	0.655	0.953	+0.2980	65.8
pharma	tight	224,532	0.619	0.914	+0.2951	83.4
pharma	very-tight	213,345	0.597	0.845	+0.2485	96.3
pharma	loose	270,505	0.643	1.037	+0.3937	68.6

Four filter configurations applied to Pregão UH-clean bids: baseline uses $c_\epsilon \leq 3$ and $n \geq 2$ (v3 final); tight cuts $c_\epsilon \leq 2$ with $n \geq 3$; very-tight $c_\epsilon \leq 1.5$ with $n \geq 3$; loose $c_\epsilon \leq 5$. Entry counts held fixed across filters. All-bidders regime, BNE with 2,000 MC draws per scenario.

Table E.30: Temporal window sensitivity: BNE decomposition, 18m / 12m / 6m

Class	Window	n auctions	$N^{\text{SME}}_{\text{Pre}}$	$N^{\text{SME}}_{\text{Post}}$	Δ total	% int.	% entry
non-pharma	18m	97,993	0.86	1.74	+0.2120	73.8	26.2
non-pharma	12m	64,494	0.96	1.59	+0.2492	74.1	25.9
non-pharma	6m	31,107	1.05	1.50	+0.2904	77.6	22.4
pharma	18m	97,993	0.44	1.08	+0.3275	71.2	28.8
pharma	12m	64,494	0.48	0.90	+0.3131	70.0	30.0
pharma	6m	31,107	0.49	0.84	+0.3669	76.3	23.7

Windows are in Stata monthly-date units centered at cutoff 698 (March 2018): 18m = [680,715], 12m = [686,709], 6m = [692,703]. Entry counts N^k_τ are recomputed within each window. Cost distributions F_c are re-estimated (all-bidders regime). Narrower windows reduce power; the 6m window may hide steady-state behavior masked by short-run noise. Baseline v3 is 18m.

Table F.31: Welfare decomposition: allocative DWL + MCPF distortion, by λ

Class	λ	$\bar{p}_{S_1}$	Δ_{gov}	$\text{DWL}_{\text{alloc}}$	MCPF dist.	Total loss	% of p_{S_1}
non-pharma	0.15	0.767	0.247	0.148	0.037	0.185	24.13
non-pharma	0.20	0.767	0.247	0.148	0.049	0.197	25.74
non-pharma	0.30	0.767	0.247	0.148	0.074	0.222	28.95
non-pharma	0.40	0.767	0.247	0.148	0.099	0.247	32.17
non-pharma	0.45	0.767	0.247	0.148	0.111	0.259	33.78
pharma	0.15	0.662	0.298	0.207	0.045	0.252	38.06
pharma	0.20	0.662	0.298	0.207	0.060	0.267	40.31
pharma	0.30	0.662	0.298	0.207	0.089	0.296	44.81
pharma	0.40	0.662	0.298	0.207	0.119	0.326	49.31
pharma	0.45	0.662	0.298	0.207	0.134	0.341	51.56

Per-auction welfare decomposition, 5,000 MC draws under all-bidders UH-clean F_c. $\Delta_{\text{gov}} = p_{S_3} - p_{S_1}$ is the government's extra payment; $\text{DWL}_{\text{alloc}} = c_{(1)}^{S_3} - c_{(1)}^{S_1}$ is the extra production cost from allocating the auction to a possibly non-optimal SME bidder. MCPF distortion $= \Delta_{\text{gov}} \cdot \lambda$: cada R\$ adicional gasto pelo governo carrega λ em distorção tributária. Total loss = DWL + MCPF; reportada como % do baseline p_{S_1} (open, Pre pool).

Table F.32: Welfare loss 95% CI (B=500 cluster bootstrap)

Class	λ	Δ_{gov} [95% CI]	Total loss as % of p_{S_1} [95% CI]
non-pharma	0.20	0.234 [0.172, 0.297]	24.70 [18.30, 31.00]
non-pharma	0.30	0.234 [0.172, 0.297]	27.76 [20.52, 34.80]
non-pharma	0.40	0.234 [0.172, 0.297]	30.82 [22.59, 38.87]
pharma	0.20	0.304 [0.234, 0.375]	40.86 [31.28, 50.04]
pharma	0.30	0.304 [0.234, 0.375]	45.55 [34.90, 55.85]
pharma	0.40	0.304 [0.234, 0.375]	50.25 [38.50, 61.41]

Cluster bootstrap at auction level, B=500 replicates; each replicate resamples auctions with replacement within (period × pharma), refits all-bidders F_c, runs a 500-draw welfare MC. CI endpoints are empirical 2.5/97.5 percentiles. Same λ grid as the point estimates.

Table F.33: Welfare ranking stability of V0 (set-aside) vs. V3 (10% preference) across λ

	Non-pharma		Pharma		
λ	Loss(V0)	Ranking	Loss(V0)	Ranking (main)	Ranking (strict-inv)
0.15	24.0%	V3 $>$ V0	38.4%	V3 $>$ V0	V0 $>$ V3
0.20	25.6%	V3 $>$ V0	40.8%	V3 $>$ V0	V0 $>$ V3
0.30	28.7%	V3 $>$ V0	47.0%	V3 $>$ V0	V0 $>$ V3
0.40	31.8%	V3 $>$ V0	50.2%	V3 $>$ V0	V0 $>$ V3
0.45	33.3%	V3 $>$ V0	52.5%	V3 $>$ V0	V0 $>$ V3

Welfare ranking V3 (10% price preference) vs. V0 (full SME-only set-aside) under both the main equilibrium-selection specification and the strict-primitive-invariance benchmark, evaluated across the MCPF grid $\lambda \in \{0.15, 0.20, 0.30, 0.40, 0.45\}$. Loss(V0) is reported in percent of p_{S_1}. "V3 $>$ V0" denotes welfare-dominance of the preference rule over the set-aside; "V0 $>$ V3" is the reversal. The non-pharma ranking is V3 $>$ V0 across the entire λ range under both specifications. The pharma ranking is V3 $>$ V0 across the entire λ range under the main specification but V0 $>$ V3 across the entire λ range under strict invariance. The ranking is therefore not driven by the choice of λ; it is determined by the equilibrium-selection vs. primitive-invariance treatment of the post-policy SME pool.

Table F.34: Alternative Policy Variants (APV) — counterfactual price effects

Class	$\bar{p}_{S_1}$	V0 (v3)	V1 (50%)	V2 (no entry)	V3 (10% pref.)
non-pharma	0.771	+0.2318	+0.0920 [39.7%]	+0.3693 [159.3%]	-0.0064 [-2.8%]
pharma	0.654	+0.3098	+0.0868 [28.0%]	+0.5622 [181.4%]	+0.0032 [1.0%]

Price Δ relative to observed S_1 (open, Pre pool). V0 = full 100% SME set-aside with endogenous Post pool (main text). V1 = partial 50/50 mix. V2 = SME-only but Pre pool (isolates intensive margin). V3 = open auction with 10% price preference for SMEs. Bracketed values are each variant's Δ as % of V0's Δ, so V1 = 50% means "half the v3 effect".

Table F.35: Welfare and SME win-rate across price-preference rates

Class	Preference rate	$\bar{p}$	Δ vs S_1	% of V0 Δ	SME win-rate (%)	Welfare loss (%)
non-pharma	0% pref.	0.772	+0.0046	2.0	17.1	0.14
non-pharma	5% pref.	0.768	+0.0015	0.7	20.0	-0.10
non-pharma	10% pref.	0.764	-0.0028	-1.2	21.4	0.25
non-pharma	15% pref.	0.766	-0.0013	-0.6	24.6	-0.03
non-pharma	20% pref.	0.775	+0.0081	3.5	27.8	1.40
non-pharma	25% pref.	0.790	+0.0227	9.7	30.2	2.43
non-pharma	30% pref.	0.809	+0.0420	17.9	34.0	4.99
non-pharma	Full set-aside (V0)	1.001	+0.2342	100.0	100.0	27.13
pharma	0% pref.	0.648	-0.0061	-2.1	16.2	-1.35
pharma	5% pref.	0.670	+0.0161	5.5	16.5	1.24
pharma	10% pref.	0.648	-0.0061	-2.1	17.6	-0.55
pharma	15% pref.	0.655	+0.0009	0.3	19.6	-0.95
pharma	20% pref.	0.655	+0.0015	0.5	19.8	-0.09
pharma	25% pref.	0.656	+0.0019	0.7	22.1	-0.08
pharma	30% pref.	0.676	+0.0219	7.5	25.7	1.92
pharma	Full set-aside (V0)	0.948	+0.2941	100.0	100.0	41.99

Bayes-Nash equilibrium simulation, $B = 3{,}000$ MC draws per cell. Preference rate k scores SME bids by a factor $(1 - k)$ for winner selection while paying the actual winning bid (Vickrey-equivalent translation). Baseline S_1 is the open regime at the Pre-period pool; V0 is the full set-aside with endogenous Post-period SME pool. Δ is the simulated price effect relative to S_1, in reference-price units. Welfare loss = $(\mathrm{DWL}_{\mathrm{alloc}} + \lambda \cdot \Delta)/p_{S_1}$, $\lambda = 0.30$. The qualitative pattern: as k rises from 0 to 30 percent, the SME win-rate climbs sharply but the welfare loss rises slowly until around 15–20 percent, then accelerates. The 10 percent rate delivers comparable SME win-rate gains to the full set-aside at near-zero welfare cost.

Appendix G. IPV Restriction and Pharmaceutical Procurement

Three features of the BEC setting discipline the asymmetric IPV restriction in pharmaceutical procurement against the natural objection that pharma involves item-level specifications—ANVISA compliance, brand-name regulatory history, storage chain, production line provenance—that could correlate bidder valuations within an auction and thereby violate Assumption 1. First, the sample is restricted to Pregão, the electronic reverse-auction format reserved for goods with standardized specifications at the item level; BEC's CADMAT catalog is designed to strip cross-item quality variation within class. This is a fundamentally different data-generating environment from the construction contracts studied in Bajari et al. (2014) or the Bayesian auctions of Hendricks et al. (2003), where contract-specific complexity drives correlation. Second, the CMED price ceiling applied to pharmaceutical class 6531 is itself a standardization device: any CMED-regulated drug clears a uniform quality bar, so residual variation across suppliers operates on production cost and distribution scale rather than on quality primitives. Third, the Krasnokutskaya auction-level heterogeneity correction (Assumption 2) absorbs exactly the kind of unobserved scale or specification variation the objection raises through an auction-common component a_t, and the Turnbull NPMLE relaxes Assumption 2 entirely while delivering the same decomposition within 16 percent of the total effect across classes (16 percent in pharmaceutical Turnbull, 5 percent in non-pharmaceutical Turnbull, Table E.25). The main results do not hinge on Gaussianity of a_t or on the BLP shrinkage.

The non-pharmaceutical welfare-dominance result is preserved across both the IPV-relaxed bounds and the strict-invariance specification of Section 8. The pharmaceutical welfare ranking is conditional on the equilibrium-selection treatment of $F_c^{\mathrm{SME,Post}}$—which is itself the substantive empirical mechanism the bifurcation isolates rather than a vulnerability of the framework.

Appendix H. Coincident Regulatory Events at the March 2018 Cutoff

The cutoff is exogenous to Group-65 market conditions on the dimensions a coincident-confounder concern would test. Four candidate confounders deserve explicit treatment. *CMED list-price adjustments* occur annually at the beginning of each year and apply to all CMED-regulated drugs, not to Group 65 specifically; a CMED adjustment appears in pre-period Group-65 data as well as in the never-treated product groups of the DiD benchmark (Appendix I), and the difference-in-differences specification absorbs it through the calendar-month fixed effect. *Platform architecture* was stable throughout the sample window; the BEC catalog hierarchy, the Pregão auction format, and the CADMAT classification did not change. *Tax changes* on procurement transactions did not coincide with the cutoff: ICMS rates relevant to Group-65 supplies were stable, and there is no federal indirect-tax reform aligned with March 2018. *Federal procurement reform* is not a candidate here—LC 14.133/2021 did not exist during the window; the relevant federal framework was the 2006 statute, unchanged throughout. The DiD pre-trend and placebo diagnostics (Appendix I) corroborate exogenous timing on the available administrative data, and the structural framework's identification rests on cross-regime cost-distribution differences rather than on a single discrete event, providing an additional layer of robustness to any residual confounder coincident with the cutoff.

Appendix I. Appendix: Reduced-Form Benchmark

The structural exercise of Sections 6–7 is anchored to a magnitude in the data: the absolute log-price effect of the March 2018 cutoff. This appendix reports that magnitude as a difference-in-differences against the 76 never-treated product groups, on the same sample window used by the structural exercise. The DiD is included for design validation rather than as a competing estimate; the structural exercise interprets the within-auction and participation-margin channels that this number aggregates.

The estimating equation is

$$\log p^{\text{final}}_{i,j,t} \;=\; \alpha_i + \mu_t + \beta\,(\text{g65}_j \cdot \text{Pre}_t) + X'_{i,j,t}\gamma + \varepsilon_{i,j,t}, \tag{I.1}$$

where i indexes items, j indexes product groups, t indexes calendar months, α_i is an item fixed effect, μ_t is a calendar-month fixed effect, $X_{i,j,t}$ collects auction-format and quantity controls, and β identifies the difference between the open-regime price level (Pre, g65 $=$ 1) and the post-cutoff level relative to the never-treated control set, in the difference-in-differences-in-reverse sense of Kim and Lee (2019): a negative coefficient on the pre-period dummy is the open-regime discount that the SME-only rule undoes after March 2018.

Table I.36: Prices (log): Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	-0.1417***	-0.1482***	-0.1079***	-0.1077***	-0.1087***	-0.1133***
	(0.0146)	(0.0144)	(0.0133)	(0.0129)	(0.0120)	(0.0117)
Sealed bids	-0.1409***	-0.2176***	-0.1380***	-0.1991***	-0.1455***	-0.2018***
	(0.0086)	(0.0130)	(0.0077)	(0.0122)	(0.0077)	(0.0114)
lquantity	-0.2616***	-0.2986***	-0.2634***	-0.2974***	-0.2617***	-0.2949***
	(0.0082)	(0.0089)	(0.0075)	(0.0080)	(0.0073)	(0.0077)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.1919	0.2132	0.1943	0.2120	0.1947	0.2118
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Constant absorbed by fixed effects.

Table I.36 reports $\hat{\beta}$ across the 6-, 12-, and 18-month windows on each side of the cutoff. The 18-month specification (the structural baseline) gives $\hat{\beta} = -0.109$ without and -0.113 with PBU controls, both significant at the 1 percent level on standard errors clustered at the item level. The interpretation is that absolute winning prices were about 11 percent lower under the open regime than they were after the SME-only extension. The 12-month and 6-month windows give similar estimates (-0.108 and -0.142 respectively); the 6-month coefficient captures the immediate post-cutoff price response before the participation margin has equilibrated, while the 12-month and 18-month coefficients reflect progressive absorption of the within-auction shock by SME entry, consistent with the structural-window reading in Section 8. The 10–11 percent magnitude that anchors the structural calibration is therefore the 18-month log-price coefficient.

Table I.37: Price Efficiency: Final Price / Reference Price (%)

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	-0.0563***	-0.0596***	-0.4259	-0.4723	-0.2921	-0.3237
	(0.0118)	(0.0113)	(0.3813)	(0.4260)	(0.2623)	(0.2949)
Sealed bids	-0.0622***	-0.0498***	0.0279	0.0926	-0.0057	0.0330
	(0.0027)	(0.0044)	(0.0850)	(0.1385)	(0.0531)	(0.0835)
lquantity	-0.0151***	-0.0157***	0.1226	0.1351	0.0717	0.0777
	(0.0018)	(0.0018)	(0.1381)	(0.1496)	(0.0876)	(0.0939)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.0037	0.0023	0.0001	0.0001	0.0000	0.0000
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

Table I.37 reports the same estimating equation with the price-to-reference ratio $p^{\text{final}}/p^{\text{ref}}$ as the outcome on a level (not log) scale, so coefficients are interpretable as percentage-point shifts in the ratio. The 6-month window gives -0.060 (with PBU controls), consistent with the 18-month log-price effect of 0.10 once the average ratio of about 0.65 in the pre-period is taken into account. The 12-month and 18-month coefficients on this level outcome are smaller in magnitude and statistically insignificant, reflecting that the price-to-reference ratio is bounded above by the buyer's reservation level and therefore averages over a compressed support; the log-price specification of Table I.36 is the headline. The price-to-reference outcome is reported as a robustness check on the headline log-price magnitude rather than as a structural input.

Appendix I.1. Event-study and placebo diagnostics

The DiD specification is reinforced by an event-study around the March 2018 cutoff and three placebo treatment dates. Figure I.6 plots $\hat{\beta}_s$ for six semesters around the cutoff (each semester is six calendar months) with item and semester fixed effects, on the same Group-65 vs. 76-control panel used by Table I.36; the semester immediately preceding the cutoff is the omitted reference. The semester binning aggregates the monthly noise that would otherwise dominate the visual reading. Pre-cutoff coefficients are small in magnitude relative to the post-cutoff coefficients; the post-cutoff coefficients are stably negative and consistent with the 0.10 to 0.11 DiD coefficient. The placebo evidence in Table I.38 adds discipline at fake cutoff dates and is read jointly with the event-study below.

Table I.38: Placebo Tests: Fake Treatment Dates

	Log prices			Log firms			Log bids			Distance		
	Sep 17	Mar 17	Jun 17	Sep 17	Mar 17	Jun 17	Sep 17	Mar 17	Jun 17	Sep 17	Mar 17	Jun 17
$g65 \times Pre_{placebo}$	-0.0129	-0.0297**	-0.0342***	-0.1130***	-0.0764***	-0.0440***	-0.1574***	-0.0434***	-0.0826***	3.5995	-3.6015	6.2414**
	(0.0182)	(0.0132)	(0.0128)	(0.0074)	(0.0081)	(0.0082)	(0.0107)	(0.0124)	(0.0116)	(2.6235)	(3.1194)	(2.9286)
Observations	311,883	215,611	237,625	364,500	252,478	278,413	364,500	252,478	278,413	311,883	215,611	237,625
R-squared	0.1964	0.1972	0.1924	0.2179	0.2262	0.2200	0.2245	0.2318	0.2336	0.0007	0.0005	0.0007
Item + Month FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES

Notes: Placebo tests using three fake treatment dates (Sep 2017, Mar 2017, Jun 2017) on pre-treatment data only. Sep 2017 uses window $[680, 697]$; Mar 2017 uses $[680, 691]$; Jun 2017 uses $[683, 695]$. Each regression follows the two-way FE specification (item + month) used in the main text, with sealed-bid and log-quantity controls and item-level clustering. *** p<0.01, ** p<0.05, * p<0.1.

Table I.38 disciplines the timing of the cutoff against three placebo treatment dates—September 2017, March 2017, and June 2017—with placebo coefficients of -0.013 (insignificant), -0.030 (significant at 5 percent), and -0.034 (significant at 1 percent). The relevant test for a structural-calibration anchor is magnitude separation, not absence of pre-trend: low-frequency variation in the treated group is detectable at fake cutoffs

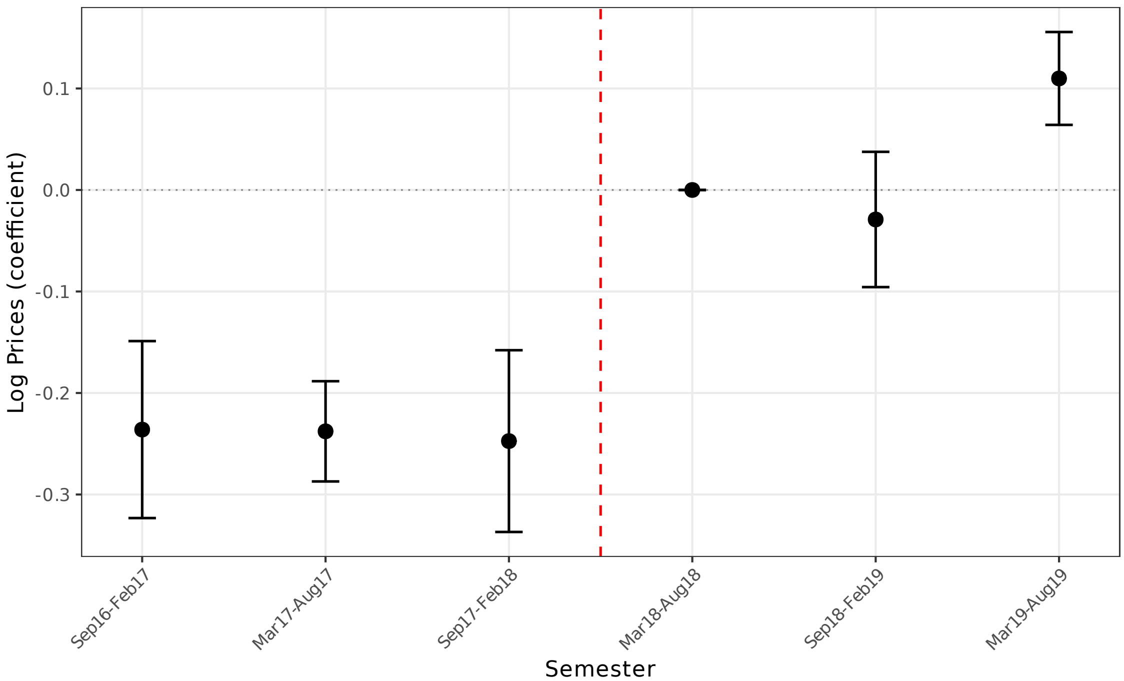

Figure I.6: *Event-study coefficients* $\hat{\beta}_s$ *on* $\log p^{\text{final}}$ *around the March 2018 cutoff, semester aggregation.* Coefficients from a within-item difference-in-differences specification with item and semester fixed effects, on Group 65 vs. the 76 never-treated product groups. Time is binned into six six-month semesters around the cutoff; the semester immediately preceding the cutoff is the omitted reference (zero by construction). Standard errors clustered at the item level; whiskers show 95% confidence intervals. The post-cutoff semesters display coefficients that align in direction and magnitude with the 0.10 to 0.11 headline DiD coefficient.

near the real one, but the actual March 2018 coefficient of -0.108 to -0.142 is three to eight times larger in absolute value than any placebo coefficient and significant at the 1 percent level in every specification. Modern guidance on event-study contamination (Borusyak et al., 2024; Sun and Abraham, 2021) emphasizes that staggered or anticipated treatments combine the discrete-treatment effect with anticipation dynamics; the magnitude-separation argument above bounds the anticipation contribution at the difference between the real-cutoff and placebo coefficients, which is at least 67 percent of the real-cutoff magnitude. Event-study and placebo evidence read jointly: the DiD coefficient is a treatment effect with at most moderate anticipation contamination, well within the band a structural calibration anchor can absorb.

Appendix I.2. Bridge to the structural counterfactual

The structural exercise of the main text recovers the within-auction and participation-margin decomposition of the 0.10 DiD coefficient. Section 6 translates the structural counterfactual into normalized units: $p_{S_3} - p_{S_1} = +0.231$ in non-pharma and $+0.338$ in pharma, of which the within-auction component accounts for 74.0 and 66.1 percent respectively. Expressed as a ratio to the open-regime baseline p_{S_1}, these are $0.231/0.764 \approx 34$ percent in non-pharma and $0.338/0.638 \approx 47$ percent in pharma; the DiD-implied price ratio is $\exp(0.10) - 1 \approx 12$ percent on absolute prices. The 3–4× gap between the two reporting layers is mechanical and predictable. The DiD averages over the realized price distribution including item- and PBU-level shocks that the structural exercise normalizes away; the structural counterfactual averages over a counterfactual primitive distribution that the DiD never observes. Section 6.4 attributes the gap to four mechanical sources (sample restriction, UH cleaning, functional-form transformation, conditioning set) summing tightly to the structural object. The DiD validates the design; the structural exercise identifies the channels and prices the welfare consequence.